



Accuracy of parameter estimation for a simple gene regulatory network model is sensitive to network motif, number of parameters estimated, and magnitude and direction of regulatory relationships

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Abstract

A gene regulatory network (GRN) is a set of transcription factors that regulate the expression of genes encoding other transcription factors. The dynamics of a GRN explain how gene expression changes over time. GRNmap is a MATLAB software package that uses ordinary differential equations to model dynamics of small-scale GRNs. We used the program to estimate production rates, expression thresholds, and regulatory weights for each transcription factor in three related literature-derived GRNs based on yeast cold shock microarray data previously collected in the Dahlquist Lab. We noticed large differences in estimated weight values when 1-2% of the expression values were missing from the input data versus when the missing values were replaced with the average value for that gene, strain, and timepoint for the 15-gene, 28-edge GRN. When the alpha value that constrains the penalty term for the least squares error function being minimized for the estimation was changed from 0.002 to 0.02 it lessened the discrepancy between parameters estimated from input with missing and no missing expression data for that GRN. However, changing the alpha value from 0.002 to 0.02 introduced more differences between estimated parameters for the two other networks (14-genes, 25-edges; 16-genes, 35 edges) that each contained a subset of the first network's genes. Due to these results, we designed a sensitivity assay that could be performed on smaller GRNs. All 21 possible "toy" networks of 3 nodes and 4 edges were generated, which fall into 11 families of network motifs. Parameters were estimated from simulated expression data created when the model underwent a forward simulation with known arbitrary weight parameters (1, 2, -1, -2), production rates, and threshold b values. Comparison of the known to estimated parameters showed that estimating production rates in addition to weights and thresholds reduced the accuracy of the results. To better understand the influence of the weight parameters, we generated all twenty-four possible weight permutations for three network motifs, bi-mutual, mutual cascade, and fan-in with two

autoregulatory loops. Through analysis of these permuted networks, we saw that the model was sensitive to the direction and magnitude of the arbitrary weight parameters assigned to networks with the same connectivity. While we would have to run more permutations to completely justify our results, through beginning to assess the biology of our networks, we can begin to understand why some weights caused more accurate results than others and also understand how network motifs may appear in larger networks and influence modeling at that level.

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Introduction

The central model of molecular biology explains how DNA is transcribed into RNA which is translated into proteins. Transcription factors bind to DNA to regulate gene expression through either activating or repressing transcription. Changes in gene expression induced by the regulatory transcription factors are the basis for all cell responses. Budding yeast, *Saccharomyces cerevisiae*, is a good model organism for studying gene regulation because it is a simple eukaryote. Many genes and regulatory pathways are conserved between yeast and humans, allowing it to be a relevant tool for understanding the fundamental biology of higher eukaryotes (Dabas et al., 2017). Additionally, databases such as the *Saccharomyces* Genome Database (SGD) and YEASTRACT, provide publicly available datasets which make analysis of transcriptional regulation in yeast more widely accessible (*Saccharomyces Genome Database | SGD*, n.d.; Teixeira et al., 2023). In budding yeast, about 250 regulatory transcription factors regulate approximately 6000 genes in the organism. Thus, each gene is regulated by approximately one to six transcription factors with each transcription factor regulating tens to thousands of genes. Together these make up the complete gene regulatory network (GRN) for yeast (Harbison et al., 2004; Lee et al., 2002; MacIsaac et al., 2006).

Previous studies have focused on determining the topology of the entire yeast GRN (Harbison et al., 2004; Lee et al., 2002; MacIsaac et al., 2006), but when considering such large systems, inferences cannot be made about network dynamics. The dynamics of small systems like the lac operon in *E. coli*, where all biochemical parameters are known, can be modeled accurately and relatively simply (Velazco et al., 2021). However, it is difficult to experimentally determine all of the biochemical parameters for systems that are larger and more interconnected. Thus, mathematical modeling is a powerful tool for understanding and making predictions

concerning network dynamics. The Dahlquist Lab has turned towards modeling small-scale networks, which are networks with about 14 to 20 genes to uncover the regulation of the cold shock response in yeast and ultimately understand how network structure and dynamics impact the system.

Previous investigators have used ordinary differential equations as the basis for modeling the dynamics of gene regulatory networks. Vu and Vohradsky (2007) used nonlinear mass balance differential equations to identify transcriptional regulators for cell cycle-regulated genes in *S. cerevisiae*. Overall, their model determined target and regulator pairs by identifying which regulator was the best predictor for the expression of each target gene. To do this, they created an algorithm in which gene expression profiles were fit to an n degree polynomial. Since Vu and Vohradsky (2007) were exploring cell cycle genes, the use of an n degree polynomial is intuitive because that would likely follow the cell cycle pattern of rhythmic expression. Candidate regulatory transcription factors were selected, and the transcription factor that minimized the error was chosen as the regulator. The error function they used is the mean square error function in which the expression profile of the target and regulator genes are compared. To minimize this error function, the procedure was repeated 100 times for each combination of regulator and target pair. There were 40 yeast cell regulator target genes and 184 potential regulators which makes 7,360 total combinations. This model assumed that the resulting change in gene expression of a target gene is based on the upstream activity of a particular regulator, linking the pattern of expression of a regulator to the expression of the target gene. Vu and Vohradsky (2007) state that a limitation of this model can be the way that candidate regulators are selected because they are selected based on the literature. For an organism like yeast, this is not as problematic because its genome is highly analyzed, but for a less studied organism where gene expression data is not as

readily available, the same methodology would not be possible. Overall, the paper serves as a good introduction to modeling with nonlinear differential equations, but it does not go on to define a full GRN, but rather stops at identifying regulators for each of the target genes. In their supplementary material, they state which genes are predicted as regulators and also state that if more than ten regulators were predicted then only the first ten are listed. They found that each target gene was regulated by as little as two to over ten of the genes from their list of 184 candidate regulators. They went on to compare their results to the YEASTRACT database and found that their identified pairs matched the literature. This paper was especially helpful to our lab because it laid the foundation for the modeling equation that we would use to design GRNmap, but overall, it seems because they stopped at regulator-target pairs, rather than defining a cell cycle GRN, the applications of their findings are not as fleshed out as they could be. It also seems that because of the limitations and the amount of data that must be available to accurately make predictions, a different modeling approach that would be more applicable to different organisms would be beneficial (Vu & Vohradsky, 2007).

Alternatively, GRNs can also be studied by “reverse engineering” a network based on *in vivo* data. Cantone et al. (2009) took a synthetic biology approach to construct a synthetic 5-gene network in *S. cerevisiae* and manipulate it for modeling purposes. The network they constructed was fully connected, with each gene regulated by other genes in the network. Galactose acted as an inducer to turn on transcription. If galactose was present in the media while culturing cells, it would bind to Gal80 which would cause downstream activation of the genes in their 5-gene network. Each promoter/transcription factor pair was chosen strategically to reduce influence on the promoters from other environmental cues. Their mathematical model was based on nonlinear differential equations and mass balance kinetics. They also had a specific model to describe the

galactose pathway which served as an activator for the entire synthetic GRN. While some simplifying assumptions were made—ignoring mRNA degradation and considering transcription and translation as one step—their model was able to predict the behavior of the network when galactose was present compared to when glucose was the carbon source. The input to the model was time series and steady-state mRNA expression data from their 5-gene network after induction with galactose. The model output was a model that simulated these gene expression changes. Specifically, the model predicted the gene expression levels of the cells in the form of mRNA and protein concentrations of the genes in the synthetic network. The purpose of this paper was to show that their *in vitro* reverse engineering and modeling assessment (IRMA) network validates the use of *in silico* mathematical models to predict *in vivo* GRN behaviors (Cantone et al., 2009).

Swain et al. (2010) compared three different commonly used methods for modeling gene regulatory networks: S-system, artificial neural networks, and general rate law of transcription. One of the methods the authors focused on is the 2001 Vohradsky Artificial Neural Network (ANN) network method (Vohradsky, 2001). This method was expanded slightly in the 2007 Vu and Vohradsky paper discussed previously but is largely similar. Thus, the conclusions from Swain et al. (2010) concerning the ANN network can be used to further evaluate the Vu and Vohradsky model. Overall, the model proposed by Vohradsky (2001) was shown to be more versatile and suitable for modeling different types of dynamics and network features, including oscillatory dynamics, which were not possible with the two other types of models that were considered. Vohradsky (2001) modeled degradation rates more accurately and also handled co-regulation much better than the other models. It seems that the other two types of models were more likely to generate unrealistic dynamics, even for small networks, which would then be

difficult to map back to the biology. Additionally, Vohradsky (2001) was more likely to yield more accurate predictions with some of the more well-studied network motifs that were more correctly modeled than others. Swain et al. (2010) explained that Vohradsky's (2001) method deserved more attention because of its reliability and likeliness to produce more accurate results for less computational power. Overall, they concluded that different modeling methods are not equivalent in terms of accuracy and interchangeability (Swain et al., 2010).

In 2015, the Dahlquist Lab published a paper detailing our lab's approach to modeling a GRN that controls the cold shock response in yeast (Dahlquist et al., 2015). The goal of this work was to use mass balance differential equations to model the dynamics of gene regulatory network in yeast regulates the cold shock response in *S. cerevisiae* (Dahlquist et al., 2015). Using a production function similar to that proposed by Vu and Vohradsky (2007) as the basis of the model, a least squares approach was taken to fit parameters to microarray data. Data from Lee et al. (2002) was used as a starting point for determining which transcription factors to include in the network. Then, the list of transcription factors of interest was narrowed by considering only those annotated as involved in the environmental stress response and also found in a dataset from Harbison et al. (2004). From there, microarray cold shock data from Schade et al. (2004) and protein half-life data from Belle et al. (2006) were used as inputs to the model. While the model is similar to that of Vu and Vohradsky (2007), rather than identifying target and regulator pairs to find which regulator was the best predictor for each target, the goal of the Dahlquist et al. (2015) model was to fit weight, threshold, and production rate parameters for each transcription factor in the network to experimental gene expression data (Dahlquist et al., 2015).

The MATLAB application of the model in the Dahlquist et al. (2015) paper was later renamed GRNmap: Gene Regulatory Network Modeling And Parameter estimation. The lab has

since collected its own microarray data from yeast subjected to cold shock for 15, 30, and 60 minutes at 13°C (Dahlquist et al., 2018), as opposed to the Schade et al. (2004) data, which had 10-, 30-, and 120-minute timepoints at 10°C. Past students normalized the raw microarray data, performed statistical analysis, and clustered the data. They identified seven different candidate GRNs—titled DB1 through DB7—relating to the cold shock response in *S. cerevisiae*. Past students also used GRNmap to estimate the parameters for these candidate networks and found that when the input gene expression data was changed slightly, the estimated parameters often varied to a larger extent than was anticipated. It is here that my work in the lab began. I first set out to determine why the estimation output varied, and answering this question led us to design a sensitivity assay using smaller 3-gene, 4-edge “toy” networks to better understand the influence of network motif, number of parameters estimated, and magnitude and direction of regulatory relationships on the accuracy of parameter estimation.

Methods

Parameter estimation with GRNmap

GRNmap is an open-source MATLAB package developed by the Dahlquist Lab to model gene regulatory networks through parameter estimation from gene expression data (Dahlquist et al., 2015). The expression of each gene in the network is modeled with a mass balance ordinary differential equation (1).

$$\frac{dx_i(t)}{dt} = \frac{P_i}{1 + \exp\left(-\sum_j (w_{ij}x_j(t) - b_j)\right)} - d_i x_i(t) \quad (1)$$

In the sigmoidal production function, P_i is mRNA production rate for gene i , d_i is the mRNA degradation rate for gene i , w is the weight term, determining the level of activation or repression

of gene j on gene i , and b is the threshold for activation or repression for each gene. It takes input mRNA expression data and degradation rate data to estimate parameters with a penalized least squares approach. Degradation rates for each gene were taken from mRNA half-life data from Neymotin et al. (2014). Cold shock yeast microarray data at 15, 30, and 60 minutes of cold shock at 13°C were collected in the Dahlquist lab (Dahlquist et al., 2018).

GRNmap can estimate all three parameters, P , b , and w , or it can hold P constant, b constant, or P and b constant to reduce the number of parameters being estimated. When all three parameters are being estimated, the total number of parameters being estimated is twice the number of genes in the network plus the number of edges.

$$E = \alpha \|\theta\|^2 + \frac{1}{Q} \sum_{\tau=1}^Q [z^d(t_r) - z^c(t_r)]^2$$

(2)

Through iteratively solving the differential equation (1) and computing the error with the error function (2), GRNmap finds the parameters that minimize the error, E . We can determine how well a model performs through calculating the ratio of the least squares error (LSE) to the minimum possible LSE, given the variation in the experimental data. The LSE:minLSE ratio accounts for different levels of variation in experimental data in different GRNs. Thus, we can use this measure to compare model results independent of the GRN.

Looking more closely at the error function (2), the function is based on the optimization parameters. The optimization parameters most integral to my work are alpha, tolfun, and tolX. TolX defines how close we want the last two parameter values of the iterates to be to each other and tolfun defines how close the least squares function should be to 0. Model iteration stops when one of the above conditions is met. The square of the parameter vector (θ) is multiplied by

alpha (α) which is used to set the regularization strength. An alpha value can be empirically determined through creating an L-curve which GRNmap has the ability to do. When running an L-curve, the goal is to determine which value of alpha will be the best balance between minimizing both the penalty term and the error term. With the L-curve function you can input values of alpha that you would like to be tested and GRNmap will iteratively run the model using the results of the output of the previous run as the input for the next run. Through plotting the penalty term against the least squares error for different choices of alpha, the alpha value best suited for the model is the one that sits in the elbow of the L-curve.

Modeling GRNs with missing expression data

Previous members of the Dahlquist Lab derived seven different small-scale candidate GRNs for *S. cerevisiae* from the YEASTRACT database, named DB1-DB7. The input Excel Workbook for each of these networks is comprised of several worksheets containing data for each gene in the network. Two worksheets define the network structure in the form of an adjacency matrix and provide initial guesses for the weight parameter for each edge (typically “1”). Several sheets contain wild type microarray data for yeast subjected to cold shock at 13°C for 15, 30, and 60 minutes, as well as data from the deletion strains ($\Delta cin5$, $\Delta gln3$, $\Delta hap4$, $\Delta hmo1$, and $\Delta zap1$). Other sheets include degradation rates from Neymotin et al. (2014) and initial guesses for the production rates (2X the degradation rate), and initial guesses for the threshold b values (typically “0”). The optimization parameters sheet stores information about the type of model run, including the alpha, tolfun, and tolX parameters. The gene expression worksheets for DB1-DB7 had missing expression datapoints for certain replicates of particular genes, strains, and timepoints because these were removed in microarray data filtering due to low expression and/or background noise. For each timepoint and gene, the expression data typically

have 3 to 5 replicates. The current version of the model, v1.10, can run when some of these expression values are missing as long as there is at least one datapoint per timepoint for a particular gene and strain. However, previous to the implementation of this feature, these missing expression values would be manually replaced with the average from the other values from that gene, strain, and timepoint. Model outputs from estimations where the missing data was manually replaced with average expression values were compared to those where these values were left missing (null). All models were run on MATLAB version 2014b on the lab computer in FEA 263.

GRNmap output workbooks can be viewed with GRNsight, a web application built by the Dahlquist Lab which automatically lays out the network graph and determines edge thickness, color, and end marker based on the estimated weight values. Nodes are colored with stripes corresponding to the average gene expression at the three cold shock timepoints. GRNsight is open source and publicly available here: <https://dondi.github.io/GRNsight/> (Dahlquist et al., 2016).

Out of the seven different candidate networks, I investigated DB1, DB2, and DB5. The missing expression value (“missing value”) input workbooks were created from copies of the workbooks already present in the Dahlquist Lab GitHub repository. For the “no missing values” workbooks, the missing expression values were replaced with the average of that timepoint for each gene and strain. P , b , and w were all estimated and model outputs were then compared to determine the impact that missing values had on parameter estimation.

Sensitivity Analysis

The results from the comparison of the outputs between the “no missing data” and “missing data” input workbooks led us to conduct a sensitivity analysis to better understand what conditions have the largest impact on parameter estimation. In this assay, we modeled all 21 possible 3-gene, 4-edge networks “toy” networks (Fig. 1). To build all possible network types, we included one or two autoregulatory edges as needed to get to four edges. We confirmed that all possible network structures were accounted for through cross-checking with current literature (Ahnert & Fink, 2016; Alon, 2019). In this assay, we started with an input workbook where all production rates were assigned the value of 0.30644, which is the average of the initial guesses of the production rates from the genes in DB5. Degradation rates were similarly assigned the average value from DB5, 0.15323, and the threshold b values were all assigned 0. Arbitrarily, each of the four weights were assigned a value of 1, -1, 2, or -2. We then used GRNmap to run a forward simulation with these parameters to generate simulated gene expression that fit these parameters.

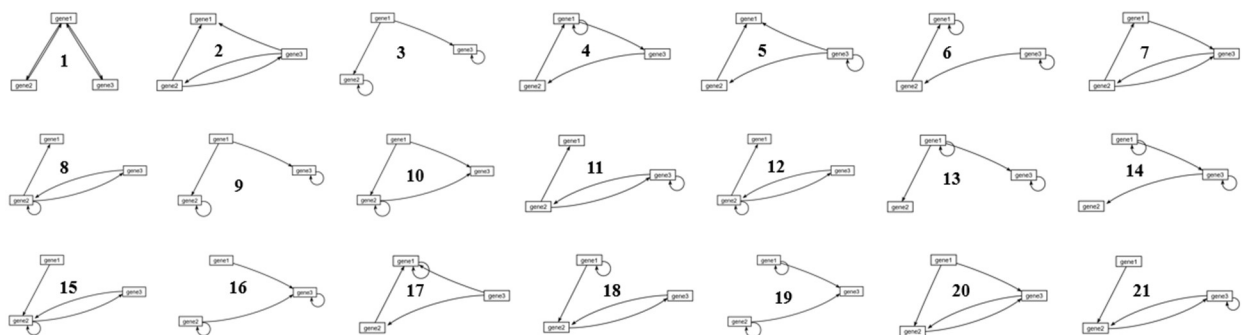


Figure 1. All 21 possible 3-gene, 4-edge networks. The number of each network corresponds to the rank in Table 13.

The bimutual 3-gene, 4-edge network was first selected to test how the optimization parameters needed to be changed to reflect the difference between using experimental data and

simulated data which inherently has no noise. Of the optimization parameters, the alpha value, tolfun, and tolX were changed. Initially, a new alpha value of 2×10^{-8} was chosen after running this network with different values for alpha (2×10^{-2} , 2×10^{-4} , 2×10^{-6} , and 2×10^{-8}). Later an L-curve was run to justify that choice of alpha. Then, different values for tolX and tolfun were also chosen. We determined that an alpha value of 2×10^{-8} and tolfun and tolX values of 2×10^{-8} were sufficient because lowering these values past this point led to overfitting. Once the optimization parameters were set, input Excel Workbooks for all twenty-one 3-gene, 4-edge networks were built, and simulated gene expression data was created with GRNmap. Each workbook had the same degradation rates, production rates, and threshold b values as the first small network, and each of the 4 edges was arbitrarily assigned a weight value of 1, -1, 2, or -2. The simulated expression data was then used to create an input file where the parameters were estimated under all possible conditions of estimating P , b , and w . Thus, for each network structure, the model was run estimating only w , estimating b and w , estimating P and w , and estimating P , b , and w .

Since the estimated model parameters were derived from simulated data, we know the ground truth for the parameters. Thus, rather than determining accuracy of the networks based on the GRNmap LSE (eq. 2), that is, the difference between the simulated and experimentally derived gene expression data, we used a weight LSE to do so. For this section of the work, the LSE does not refer to the error equation in (2), but rather the following equation (3) below.

$$LSE = \sum_{i=1}^n (w_i - w'_i)^2$$

(3)

Where n represents the number of edges, w_i represents the true weight, and w'_i represents the observed weight.

Additionally, because the four different weight values were assigned arbitrarily, there were other possible weight assignments for each network structure that could be generated by permuting the four weights assigned to each edge. Thus, for each of the 21 network structures, there were 4! or 24 possible networks that could have been generated from assigning different weights to edges. To investigate the influence of the weight value assignment on the model results, parameters were estimated for all-possible permutations of the weight values for three selected networks: bi-mutual, mutual cascade, and fan-in with two autoregulatory loops.

Automating network workbook creation

As seen in the previous sections, there were many different input workbooks that needed to be created that were often mostly copies from each other with a few cells changed. Because the process to create these input workbooks was quite time consuming and error prone, I created an R script to automate this process. The script is available on the Dahlquist Lab GitHub repository

(<https://github.com/kdahlquist/DahlquistLab/tree/50e6c8c8817efe5daba60a911e3855cc85724b2b/data/Spring2026/Automating-toy-network-file-making-process/Rscripts>) and was used to create workbooks for one of the three permutation experiments that were run.

Results

Analyzing the consequences of missing gene expression datapoints

GRNmap (v1.10) can perform parameter estimation when there are missing gene expression data values from input the Excel Workbook as long as there is at least one value per gene, strain, and timepoint. However, in previous versions of the code, missing values had to be

manually replaced with the average value for that strain and timepoint. Filling in missing values with the average value does not change the average value for that gene and timepoint and should not have changed the parameter estimation results. When we compared the model results from input that contained missing expression values and input where the missing values were replaced with the average value for that gene, strain, and timepoint, we noticed discrepancies in the optimized weights, production rates, and threshold b values. Thus, we chose to further investigate the extent of these discrepancies.

Table 1. Number of missing gene expression values in the input workbooks for DB5, DB1, and DB2 per gene, strain, and timepoint, ordered alphabetically by gene. *Indicates that this gene should have had a missing value but the input workbook that was run had already been manually filled with the average value. NA indicates that the gene was not present in the network.

gene	strain	timepoint	number of missing values DB5	number of missing values DB1	number of missing values DB2
ACE2	wild type	30	1	1	NA
ACE2	$\Delta cin5$	30	1	1	NA
AFT2	wild type	30	NA	1	NA
AFT2	$\Delta cin5$	60	NA	1	NA
AFT2	$\Delta zap1$	60	1*	2	NA
CIN5	$\Delta cin5$	15	1	1	1
CIN5	$\Delta cin5$	60	1	1	1
CIN5	$\Delta zap1$	15	1	1	NA
GCN4	$\Delta cin5$	60	NA	1	NA
GCR2	$\Delta gln3$	30	1	NA	1
GLN3	$\Delta cin5$	60	1*	1	1*
MSN2	$\Delta cin5$	15	1	1	1
SFP1	$\Delta cin5$	60	1*	1	1*
STB5	$\Delta hmo1$	30	1	NA	1
STB5	$\Delta gln3$	15	1	NA	1
STB5	$\Delta hap4$	15	1	NA	1
STB5	$\Delta hap4$	30	1	NA	1
STB5	$\Delta hap4$	60	1	NA	1
STB5	$\Delta hmo1$	15	2	NA	2
STB5	$\Delta hmo1$	60	2	NA	2
SWI4	$\Delta cin5$	15	1*	1	1
SWI4	$\Delta hap4$	30	1	1	1
SWI5	$\Delta gln3$	60	1	NA	1
SWI5	$\Delta hap4$	15	1	NA	1
YOX1	$\Delta gln3$	15	1	1	1
ZAP1	wild type	60	1	1	NA
ZAP1	$\Delta gln3$	15	1	1	NA
ZAP1	$\Delta gln3$	30	1	1	NA
ZAP1	$\Delta gln3$	60	1	1	NA
ZAP1	$\Delta hap4$	15	1	1	NA

ZAP1	$\Delta zap1$	15	1	1	NA
ZAP1	$\Delta zap1$	60	1	1	NA

We first explored DB5, a 15-node and 28-edge network (Appendix A Fig. 1) as the basis for our analysis. Missing values occurred in both the wild type expression data and each of the deletion strains' data: $\Delta cin5$, $\Delta gln3$, $\Delta hap4$, $\Delta hmo1$, and $\Delta zap1$. In those input worksheets there were two, four, seven, six, five, and three missing values, respectively, as seen in Table 1. DB5 has a total of 1,095 expression datapoints, with only 27 (2.47%) missing values. Later we also analyzed DB1—16 genes and 36 edges—and DB2—14 genes and 25 edges—both of which shared genes with DB5. Note that DB2 does not have the ZAP1 gene in the network and therefore did not have the expression data for the $\Delta zap1$ strain. DB1 has a total of 1,168 expression datapoints with 23 (1.97%) of the datapoints missing and DB2 has a total of 854 expression datapoints with 18 (2.11%) missing.

As seen in the top row of Appendix A Fig. 1 and in Table 2, the sign of the edge weight values which indicate activation or repression is inconsistent for ten of the 28 edges in the network when comparing model output from an input workbook with missing expression values to an input workbook where the missing expression values were replaced with the average value for that gene and timepoint. There are also clear differences in weight magnitudes. This is surprising because while it is expected that there might be some slight differences between the estimated weight parameters between these two conditions, the interpretation that genes which might be repressors have changed to activators or vice versa leaves us to wonder how well the model is capturing the true regulatory relationships.

Table 2. Optimized weight parameters for DB5 from input workbooks where expression values were missing (“missing”) versus input workbooks where the missing values were replaced by the average value for that gene and timepoint (“no missing”) when the alpha value was 0.002 or 0.02. Dark blue cells correspond to the largest magnitude negative weights, and dark red cells correspond to largest magnitude positive weights. White cells correspond to a weight value close to zero. “Absolute difference” is the absolute value of the difference between the weights for “missing” values and “no missing” values for alpha of 0.002 and 0.02. The rank for each alpha was determined by sorting the weights based on absolute difference. A small rank corresponds to a small absolute difference.

Regulator	Target	w when α value = 0.002 (missing)	w when α value = 0.002 (no missing)	w when α value = 0.02 (missing)	w when α value = 0.02 (no missing)	Absolute difference (missing - no missing) α value = 0.002	Absolute difference (missing - no missing) α value = 0.02	Rank α = 0.002	Rank α = 0.02
ACE2	ASH1	-1.167	-0.926	-1.013	-0.818	0.241	0.195	7	9
ASH1	YHP1	-1.573	0.106	-1.829	-1.165	1.679	0.664	19	14
CIN5	HAP4	0.280	0.478	0.017	0.186	0.198	0.169	4	8
CIN5	SFP1	-0.074	0.329	-0.043	0.225	0.402	0.268	9	10
CIN5	STB5	-0.443	-0.240	-0.227	-0.298	0.203	0.072	5	5
CIN5	YHP1	0.284	0.079	0.198	0.204	0.205	0.006	6	2
GCR2	MSN2	-3.388	3.251	-2.237	1.875	6.638	4.112	26	27
HMO1	CIN5	0.826	0.395	0.653	0.585	0.431	0.068	10	4
HMO1	HAP4	-0.737	0.467	-0.458	0.440	1.204	0.898	16	17
HMO1	HMO1	0.670	0.545	0.765	0.618	0.125	0.147	2	7
HMO1	MSN2	0.221	0.101	0.217	0.142	0.120	0.074	1	6
HMO1	YOX1	0.556	0.048	0.555	0.100	0.508	0.455	11	12
MSN2	ASH1	-2.444	-1.530	-1.527	-1.219	0.914	0.308	12	11
MSN2	CIN5	-1.485	0.707	-1.049	0.167	2.193	1.216	21	21
MSN2	HAP4	2.987	-4.002	1.769	-2.664	6.990	4.433	28	28
MSN2	SFP1	-0.124	-1.268	-0.006	-1.025	1.145	1.019	15	19
MSN2	SWI4	-0.866	0.440	-0.951	-0.286	1.306	0.665	18	15
MSN2	YHP1	0.089	2.894	-0.511	1.044	2.806	1.555	23	22
MSN2	YOX1	-1.983	1.948	-1.524	1.036	3.931	2.560	25	25
SFP1	SWI5	-2.850	-1.793	-2.016	-1.447	1.057	0.569	14	13
STB5	HAP4	0.774	1.704	-1.146	-0.053	0.930	1.093	13	20
STB5	SFP1	-0.549	1.533	-0.043	0.637	2.082	0.680	20	16
SWI4	HAP4	-2.953	-0.292	-1.619	0.486	2.660	2.105	22	24
SWI4	YHP1	-0.023	-3.029	0.003	-0.986	3.006	0.990	24	18
SWI4	YOX1	-0.260	-1.541	0.342	-1.269	1.281	1.611	17	23
SWI5	ASH1	5.212	-1.771	2.865	-1.122	6.984	3.987	27	26
YHP1	GLN3	1.518	1.806	0.746	0.749	0.288	0.003	8	1
ZAP1	ACE2	0.772	0.924	0.882	0.938	0.152	0.056	3	3

As seen in Table 2, when considering just the data for when the alpha value was 0.002, we can evaluate the variation in weight parameter estimation through first considering the absolute value of the difference between weights when the input workbooks included missing

expression values and when missing values in the input workbooks were replaced with the average value for that gene and timepoint. The maximum difference is 6.990 and the minimum difference is 0.120. The average difference is 1.774 and the standard deviation is 2.062. To put this in perspective, the range of weight values for the “missing data” condition is from -3.388 to 5.212 (8.600 magnitude difference) and -4.002 to 3.251 (7.253 magnitude difference). There are three weights whose difference is above 6.5 with the next largest weight after those values being 3.931. The three edges with the largest weight differences—from largest to smallest—are MSN2 to HAP4, SWI5 to ASH1, and GCR2 to MSN2. There are eight edges whose weight difference is less than 5% of the largest difference value, so 20 of the 28 weights have differences above that 5%. Overall, the sum of the absolute value of the difference for all the weights is 49.679 so the three largest differences in weights account for 41.49% of the total difference between the “missing” and “no missing” value workbooks. As can be seen on the heat map on overlaid on Table 2, when weights change sign between “missing” and “no missing” input workbooks, each of these large difference weight parameters all changed sign.

Production rates (Table 3) and threshold b values (Table 4) were also estimated, and we investigated whether the large differences in the edge weight values corresponded to differences in P and/or b for the regulator and target genes for those edges.

Table 3. Optimized production rates for DB5 from input workbooks where expression values were missing (“missing”) versus input workbooks where the missing values were replaced by the average value for that gene and timepoint (“no missing”) when the alpha value was 0.002 as compared to an alpha value of 0.02. “Absolute difference” is the absolute value of the difference between the production rates for “missing” values and “no missing” values for alpha of 0.002 and 0.02. Cells with darker green indicate a larger difference in magnitude and lighter green indicates a smaller difference in magnitude.

Gene ID	P when α value = 0.002 (missing)	P when α value = 0.002 (no missing)	P when α value = 0.02 (missing)	P when α value = 0.02 (no missing)	Absolute difference (missing - no missing) α value = 0.002	Absolute difference (missing - no missing) α value = 0.02
ACE2	0.2017	0.1632	0.1700	0.1640	0.0385	0.006
ASH1	1.6768	2.0211	0.9727	1.4602	0.3443	0.4875
CIN5	0.6556	0.9489	0.8277	0.5510	0.2933	0.2767
GCR2	0.2315	0.2353	0.2468	0.2540	0.0038	0.0072
GLN3	0.3021	0.2878	0.3496	0.3515	0.0143	0.0019
HAP4	1.3023	1.6615	0.8265	1.0899	0.3592	0.2634
HMO1	0.3067	0.2387	0.2369	0.2019	0.068	0.035
MSN2	2.5566	0.8858	1.5993	0.8419	1.6708	0.7574
SFP1	1.5545	1.2055	0.9980	1.1909	0.349	0.1929
STB5	0.1197	0.3193	0.2452	0.2312	0.1996	0.014
SWI4	0.3158	0.2419	0.6123	0.3304	0.0739	0.2819
SWI5	1.9213	1.2455	1.4052	0.9614	0.6758	0.4438
YHP1	0.2080	0.3431	0.5486	0.2545	0.1351	0.2941
YOX1	1.3912	0.8609	0.8984	0.6760	0.5303	0.2224
ZAP1	0.1282	0.1298	0.1291	0.1305	0.0016	0.0014

Table 4. Optimized threshold b values for DB5 from input workbooks where expression values were missing (“missing”) versus input workbooks where the missing values were replaced by the average value for that gene and timepoint (“no missing”) when the alpha value was 0.002 as compared to an alpha value of 0.02. “Absolute difference” is the absolute value of the difference between the production rates for “missing” values and “no missing” values for alpha of 0.002 and 0.02. Cells with darker green indicate a larger difference in magnitude and lighter green indicates a smaller difference in magnitude.

Gene ID	b when α value = 0.002 (missing)	b when α value = 0.002 (no missing)	b when α value = 0.02 (missing)	b when α value = 0.02 (no missing)	Absolute difference (missing - no missing) α value = 0.002	Absolute difference (missing - no missing) α value = 0.02
ACE2	0.8985	0.6004	0.6113	0.6066	0.2981	0.0047
ASH1	2.5823	-1.7729	1.2075	-1.1212	4.3552	2.3287
CIN5	-0.4781	2.3615	0.2560	1.0159	2.8396	0.7599
GCR2	0	0	0	0	0	0
GLN3	0.6939	0.8206	0.4153	0.4408	0.1267	0.0255
HAP4	3.6112	0.7628	0.4814	-0.0176	2.8484	0.499
HMO1	1.6462	0.9676	1.3148	0.6887	0.6786	0.6261
MSN2	-1.4615	4.8043	-0.8209	3.2653	6.2658	4.0862
SFP1	-0.2939	1.0500	-0.3683	0.0307	1.3439	0.399
STB5	-1.5957	0.5720	0.2296	-0.0337	2.1677	0.2633
SWI4	-0.9019	0.3354	-0.0567	-0.0194	1.2373	0.0373
SWI5	-1.7844	-0.5902	-0.8953	-0.3894	1.1942	0.5059
YHP1	-0.3135	2.2864	0.0787	0.7576	2.5999	0.6789
YOX1	-0.4477	2.0755	0.1919	0.8898	2.5232	0.6979
ZAP1	0	0	0	0	0	0

The absolute value of the difference in production rates was the highest magnitude for MSN2 and SWI5, followed by YOX1, HAP4, SFP1, and ASH1. We were curious to see how the discrepancies in production rates were related to discrepancies in optimized weights. MSN2 regulates seven genes (out-degree of seven), which is the highest number of all genes in this network. Additionally, it has an in-degree of two. On the other hand, SWI5 has an out-degree of one and an in-degree of one. Considering the optimized weight results, we see that MSN2 and SWI5 are regulators for the two edge weights that have the largest magnitude difference between the “missing” and “no missing” value inputs while HAP4 and ASH1 are their targets, respectively. MSN2 is also the target for the third largest edge weight difference.

Similarly for the threshold b values, we were curious to see if there was a relationship between variation in threshold b values and variation in P and w when the input workbooks had “missing” versus “no missing” expression values. MSN2, ASH1, HAP4, CIN5, YHP1, YOX1, and STB5 had the largest absolute values of the difference in b , in that order, and all were greater than 2. Of these, MSN2, YOX1, HAP4, and ASH1 also had large differences in estimated production rates, and MSN2, HAP4, and ASH1 were also involved in the edges with the largest differences in weights, either as regulator or target. Since GRNmap is estimating P , b , and w as a system at the same time, it makes sense that there would be discrepancies in the edge weights, production rates, and threshold b values for the same genes. Differences in one parameter would compensate for differences in the other parameters, while still achieving a good fit between the simulated and experimental expression data. We wondered if genes with a larger in- or out-degree were more likely to have inconsistencies in outputs. While MSN2 did have a large difference with an out-degree of seven, if out-degree was the sole influence on whether the parameter outputs were different, we would have also expected HMO1 with an out-degree of five to have had a large difference in parameter values, but it has one of the smaller differences as seen in Tables 2 and 3. However, HMO1 has an in-degree of 1, which is an autoregulatory edge, so P and b completely determine its expression. Overall, when we look at the differences between parameter output between the “missing” and “no missing” data input workbooks, we see that generally that genes with larger magnitude differences in parameter values have larger differences in all three parameters and that genes with lower magnitude differences have smaller differences in all three parameters.

Because the inconsistencies were more extreme than expected, especially when we consider that less than 3% of the expression datapoints were missing, optimization parameters

were investigated to determine if they were a contributing factor. Inspection of the input workbooks inherited from prior students in the lab showed that the alpha value was 0.002. Figure 2 shows the L-curve that was the basis for this choice (Klein, 2018), and the value of 0.002 is not exactly in the elbow of the curve. So, we suspected that a larger alpha value of 0.02 might be a better choice. Thus, the model estimation was performed again on the DB5 input workbooks with “missing data” and “no missing data” with alpha 0.02.

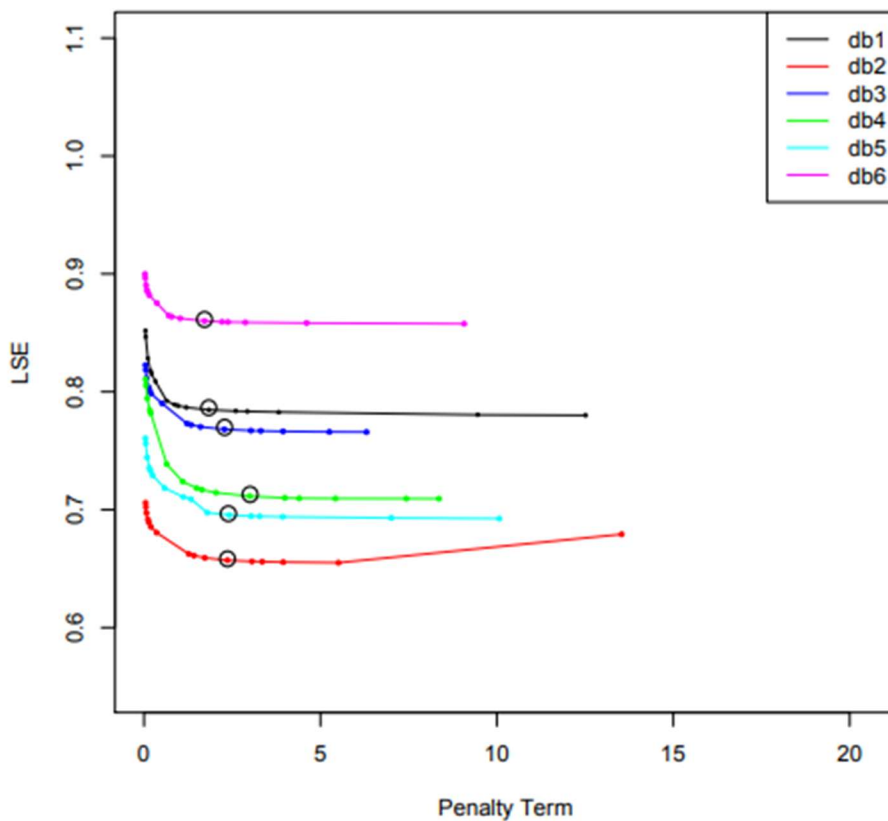


Figure 2. “L-curve plots of least squares error (LSE) versus penalty term value for db1-db6 show efficient regularization at $\alpha=0.002$ (corresponding runs are circled in black). Individual points correspond to model runs conducted with unique α values, which ranged from 0.8000-0.0001. Sixteen runs were completed for each network. The L-curves are color coded based on GRN number as indicated in the legend. The L-Curves.R script was used to generate this graph.” (Klein, 2018).

Looking at the data (Appendix A Fig.1, bottom row and Table 2), 12 estimated weight parameters changed sign between the “missing” and “no missing” data at alpha of 0.02, while

only 10 changed signs with alpha 0.002. While there were more sign changes for the larger alpha value, the magnitude of these sign changes was smaller resulting in a smaller difference in the outputs. When alpha was 0.02, the maximum difference for the weights was 4.433, and the minimum difference was 0.003. The average difference was 1.07 with a standard deviation of 1.28. To put this in perspective, the range of weight values for the “missing data” condition is from -2.237 to 2.865 (5.102 magnitude difference) and -2.664 to 1.875 (4.539 magnitude difference). There were nine weights which were less than 5% of the maximum weight, so 19 of the 28 weights were above that threshold. Comparing the results of the estimation with alpha 0.02 to those when the alpha was 0.002, we see that the average difference in weight values decreased by 40% and the standard deviation also decreased. However, the range of weight values also decreased by 40%, which is expected because of the dampening effect alpha has on the parameter estimates of the least squares error function. Thus, the smaller differences in weight values might simply be explained by the change in alpha value. Eight of the 12 weight values that changed sign at alpha 0.02 are the same as those that changed sign at alpha 0.002. Of the weights that changed sign, none of them had a difference that was below 5%. The three edges with the largest weight differences weights—from largest to smallest—are still MSN2 to HAP4, SWI5 to ASH1, and GCR2 to MSN2. Moreover, of the eleven edge weights that changed sign, three were regulated by SWI4, three were regulated by MSN2, and the remaining five were regulated by HMO1, STB5, CIN5, GRC2, and SWI5, respectively.

Tables 3 and 4 show how these weight differences were related to differences in estimated values for production rates and threshold b . The mean difference in production rates for alpha 0.002 was 0.3172 and the standard deviation was 0.4264, as compared to when the alpha value was 0.02 where the mean difference was 0.2190 and the standard deviation was

0.2218. Production rates that had the largest difference when the alpha value was 0.002, had a reduced mean difference when the alpha value was changed to 0.02. Overall, the total magnitude of the mean difference decreased with an alpha value of 0.02, but like with the edge weight values, this is likely due to an overall decrease in the range of production rate values. Four of the top six genes with the largest differences in production rates when alpha was 0.002, were the largest when alpha was 0.02: MSN2, SWI5, HAP4, ASH1. For the threshold b values (Table 4) when the alpha value was 0.002, the mean difference was 1.8986 and the standard deviation was 1.7625, as compared to when the alpha value was 0.02, the mean difference was 0.7275 and the standard deviation was 1.0966, again likely explained by an overall decrease in the range of threshold b values. Similar to the production rates, the optimized threshold b values with the largest magnitude differences decreased with an alpha value of 0.02. The genes with the largest differences between input workbooks with “missing” versus “no missing” expression values were MSN2 and ASH1 for both alpha values, however with an alpha of 0.02, the differences did decrease by about 2 units for each gene. Five of the top six genes with the largest differences in threshold b values when alpha was 0.002, were the largest when alpha was 0.02: MSN2, ASH1, CIN5, YHP1, and YOX1. In summary, there were quite a few differences in estimated parameter values with input workbooks with “missing” expression data versus “no missing” expression data for DB5, whether the alpha value was 0.002 or 0.02. The genes that varied the most across all three estimated parameters were: MSN2, HAP4, ASH1, YOX1, and to a lesser extent SWI5 and YHP1.

We next performed parameter estimation for two additional gene regulatory networks, DB1 (Appendix A Figure 2, Tables 5, 6, and 7) and DB2 (Appendix A Figure 3, Tables 8, 9, and 10), under the four conditions of “no missing” expression values, “missing” values at alpha

0.002 and 0.02. For DB1, when the alpha value was 0.002, there was only one weight that changed sign between input workbooks that had “no missing” expression data and “missing” data (Appendix A Figure 2, top row; Table 5): HMO1 to CIN5. That weight change is not meaningfully different because the magnitude of the weight is very small (0.031 vs. -0.092), about 5% of the magnitude of the largest weight. The edge weight of MSN2 to AFT2 had a very large change in magnitude (5.943 to 0.029), although it did not change sign. The next largest difference was 0.834 for MSN2 to CIN5. The minimum difference was 0.002 for HMO1 to GCN4. Overall, the average of the weight differences was 0.301 with a standard deviation of 0.981 for alpha 0.002 as compared to weight ranges of -3.357 to 5.943 (magnitude 9.300; “missing” data) and -3.393 to 2.065 (magnitude 5.458; “no missing” data). While the weight ranges were similar to those of DB5, the differences were far smaller between the “missing” data and “no missing” data conditions.

Then, when alpha value was changed to 0.02, all of the weights from the two parameter estimations (“missing” vs. “no missing” input expression data) kept the same sign (Appendix A Figure 2, top row; Table 5). Overall, the average of the weight differences was 0.028 with a standard deviation of 0.049 as compared to weight ranges of -2.183 to 2.026 (magnitude 4.209) and -2.196 to 2.123 (magnitude 4.319). The maximum edge weight difference was 0.252 (MSN2 to CIN5) and the minimum value was 7.61×10^{-5} (HMO1 to GCN4). A similar pattern to DB5 of smaller weight parameter ranges when alpha was 0.02 versus 0.002 was observed.

Table 5. Optimized weight parameters for DB1 from input workbooks where expression values were missing (“missing”) versus input workbooks where the missing values were replaced by the average value for that gene and timepoint (“no missing”) when the alpha value was 0.002 or 0.02. Dark blue cells correspond to the largest magnitude negative weights, and dark red cells correspond to largest magnitude positive weights. White cells correspond to a weight value close to zero. “Absolute difference” is the absolute value of the difference between the weights for “missing” values and “no missing” values for alpha of 0.002 and 0.02. The rank for each alpha was determined by sorting the weights based on absolute difference. A small rank corresponds to a small absolute difference.

Regulator	Target	w when α value = 0.002 (missing)	w when α value = 0.002 (no missing)	w when α value = 0.02 (missing)	w when α value = 0.02 (no missing)	Absolute difference (missing - no missing) α value = 0.002	Absolute difference (missing - no missing) α value = 0.02	Rank α = 0.002	Rank α = 0.02
ABF1	ASF1	2.043	2.020	1.016	1.011	0.023	0.004	8	9
ABF1	ASH1	1.370	1.228	1.831	1.838	0.141	0.007	23	15
ABF1	HMO1	-0.137	-0.900	-0.963	-1.081	0.764	0.118	34	35
ABF1	SFP1	1.560	1.546	0.495	0.476	0.014	0.019	7	24
ABF1	SWI4	-3.357	-2.880	-1.423	-1.340	0.477	0.083	33	33
ACE2	ASH1	-1.319	-1.316	-1.101	-1.107	0.004	0.005	2	11
ASH1	YHP1	-2.670	-2.823	-1.792	-1.792	0.154	0.000	25	2
CIN5	ASF1	-0.137	-0.141	-0.141	-0.148	0.004	0.007	4	13
CIN5	HAP4	0.216	0.212	0.198	0.201	0.004	0.002	3	6
CIN5	SFP1	0.013	0.003	-0.025	-0.038	0.009	0.013	6	22
CIN5	YHP1	0.356	0.400	0.166	0.167	0.044	0.002	13	5
GCN4	GLN3	0.438	0.395	-0.580	-0.586	0.044	0.006	12	12
GLN3	GCN4	0.788	0.782	0.911	0.910	0.006	0.001	5	3
HAP4	GCN4	0.331	0.305	0.175	0.162	0.026	0.012	9	21
HMO1	ASF1	0.032	0.079	0.106	0.101	0.047	0.005	15	10
HMO1	CIN5	0.031	-0.092	0.707	0.743	0.123	0.036	21	30
HMO1	GCN4	-0.008	-0.010	-0.119	-0.119	0.002	0.000	1	1
HMO1	HAP4	0.909	0.758	-0.378	-0.405	0.151	0.027	24	29
HMO1	HMO1	0.618	0.456	0.606	0.609	0.161	0.004	27	8
HMO1	MSN2	0.509	0.277	2.026	2.123	0.232	0.097	31	34
HMO1	YOX1	0.235	0.281	0.620	0.639	0.046	0.019	14	23
MSN2	AFT2	5.943	0.029	1.249	1.236	5.914	0.012	36	19
MSN2	ASF1	0.269	0.133	-0.066	-0.058	0.135	0.008	22	16
MSN2	ASH1	-0.770	-0.984	-0.887	-0.878	0.214	0.010	29	17
MSN2	CIN5	1.231	2.065	-0.850	-1.102	0.834	0.252	35	36
MSN2	HAP4	-3.165	-3.393	0.823	0.900	0.228	0.078	30	32
MSN2	SFP1	0.436	0.521	0.558	0.577	0.085	0.020	18	25
MSN2	SWI4	-0.965	-0.912	-0.615	-0.617	0.053	0.002	16	4
MSN2	YHP1	0.586	0.828	0.031	0.039	0.242	0.007	32	14
MSN2	YOX1	0.339	0.168	-0.916	-0.992	0.171	0.076	28	31
SWI4	HAP4	-2.549	-2.493	-2.183	-2.196	0.056	0.012	17	20
SWI4	YHP1	1.259	1.099	0.280	0.270	0.160	0.010	26	18
SWI4	YOX1	1.484	1.594	0.662	0.685	0.109	0.023	20	27
YHP1	ASF1	-0.270	-0.233	-0.421	-0.396	0.037	0.025	11	28
YHP1	GLN3	1.282	1.252	0.786	0.784	0.031	0.002	10	7
ZAP1	ACE2	1.225	1.320	0.935	0.914	0.095	0.021	19	26

Additionally, when the alpha value was changed from 0.002 to 0.02, the differences were reduced for the weights that previously had large differences (Table 5). Of the 36 weights, 34 had reduced differences. Additionally, for alpha 0.02, all weights except for the two that had the largest magnitude differences only differed in the hundredths place. When the alpha value was 0.002, the four edge weights with the largest magnitude difference between “missing” and “no missing” values were MSN2 to AFT2, MSN2 to CIN5, ABF1 to HMO1, and ABF1 to SWI4, respectively. When the alpha value was 0.02, three of the top four edge weights with the largest magnitude difference between “missing” and “no missing” values were in the top four for alpha 0.02: MSN2 to CIN5, ABF1 to HMO1, HMO1 to MSN2, and ABF1 to SWI4, respectively. As with the analysis of DB5 results, we examined the estimated production rates and threshold b values that corresponded to genes with the largest differences in weight values.

Table 6. Optimized production rates for DB1 from input workbooks where expression values were missing (“missing”) versus input workbooks where the missing values were replaced by the average value for that gene and timepoint (“no missing”) when the alpha value was 0.002 as compared to an alpha value of 0.02. “Absolute difference” is the absolute value of the difference between the production rates for “missing” values and “no missing” values for alpha of 0.002 and 0.02. Cells with darker green indicate a larger difference in magnitude and lighter green indicates a smaller difference in magnitude.

Gene ID	P when α value = 0.002 (missing)	P when α value = 0.002 (no missing)	P when α value = 0.02 (missing)	P when α value = 0.02 (no missing)	Absolute difference (missing - no missing) α value = 0.002	Absolute difference (missing - no missing) α value = 0.02
ABF1	0.0576	0.0577	0.0570	0.0570	8.558E-05	6.262E-06
ACE2	0.1337	0.1255	0.1648	0.1627	8.132E-03	2.115E-03
AFT2	0.2900	0.4781	0.4222	0.4167	0.188	5.484E-03
ASF1	0.1582	0.1583	0.1659	0.1640	6.201E-05	1.925E-03
ASH1	1.5320	1.6212	0.9379	0.9304	0.089	7.462E-03
CIN5	1.1530	1.2399	0.7499	0.8374	0.087	0.087
GCN4	0.1811	0.1741	0.0676	0.0674	6.965E-03	2.749E-04
GLN3	0.3602	0.3612	0.4225	0.4236	1.041E-03	1.110E-03
HAP4	2.3414	2.4381	0.8172	0.8227	0.097	5.499E-03
HMO1	0.4829	0.4640	0.4354	0.4272	0.019	8.137E-03
MSN2	0.5751	1.0591	0.3031	0.3013	0.484	1.774E-03
SFP1	1.0068	1.0270	0.7986	0.7989	0.020	3.144E-04
SWI4	0.2280	0.2496	0.6914	0.6793	0.022	0.012
YHP1	0.1513	0.1462	0.3063	0.3046	5.053E-03	1.688E-03
YOX1	0.7854	0.8283	0.5486	0.5545	0.043	5.906E-03
ZAP1	0.1265	0.1312	0.1275	0.1321	4.678E-03	4.638E-03

Table 7. Optimized threshold b values for DB1 from input workbooks where expression values were missing (“missing”) versus input workbooks where the missing values were replaced by the average value for that gene and timepoint (“no missing”) when the alpha value was 0.002 as compared to an alpha value of 0.02. “Absolute difference” is the absolute value of the difference between the production rates for “missing” values and “no missing” values for alpha of 0.002 and 0.02. Cells with darker green indicate a larger difference in magnitude and lighter green indicates a smaller difference in magnitude.

Gene ID	b when α value = 0.002 (missing)	b when α value = 0.002 (no missing)	b when α value = 0.02 (missing)	b when α value = 0.02 (no missing)	Absolute difference (missing - no missing) α value = 0.002	Absolute difference (missing - no missing) α value = 0.02
ABF1	0	0	0	0	0	0
ACE2	0.2597	0.1494	0.5855	0.5706	0.1103	0.0149
AFT2	4.5173	-0.1113	1.1618	1.1290	4.6286	0.0328
ASF1	1.3602	1.2858	0.4527	0.4536	0.0744	0.0009
ASH1	0.7661	0.4809	0.4555	0.4678	0.2852	0.0123
CIN5	2.7004	3.7456	0.4078	0.3814	1.0452	0.0264
GCN4	2.4220	2.3435	0.6573	0.6361	0.0785	0.0212
GLN3	1.3442	1.2861	0.3638	0.3615	0.0581	0.0023
HAP4	-1.7464	-2.2561	0.5132	0.5277	0.5097	0.0145
HMO1	1.3118	0.5214	0.6186	0.5147	0.7904	0.1039
MSN2	0.8992	1.4977	-0.2309	-0.2128	0.5985	0.0181
SFP1	1.2437	1.3742	0.2344	0.2220	0.1305	0.0124
SWI4	-3.3894	-2.8856	-0.1346	-0.1293	0.5038	0.0053
YHP1	0.3096	0.3165	0.2876	0.2768	0.0069	0.0108
YOX1	2.9861	3.0503	0.6044	0.6119	0.0642	0.0075
ZAP1	0	0	0	0	0	0

With regard to production rates (Table 6), when the alpha value was 0.002, the mean difference between estimation when there were “missing” values versus “no missing” values was 0.0672 and the standard deviation was 0.1226. When the alpha value was 0.02, the mean difference was 0.0091 and the standard deviation was 0.0212. Differences for 13 of the 16 production rates were reduced and one stayed the same. Additionally, when the alpha was increased to 0.02, the largest difference in estimated production rates is 0.087 which is relatively small (9.3% of the largest magnitude production rate at alpha 0.02). For comparison, at alpha 0.002, the largest difference in production rates was 0.484, which is 20% of the largest

production rate. As with DB5, the range of values for the estimated production rates was smaller at alpha 0.02 than alpha 0.002, explaining this overall pattern. With regard to individual genes, at alpha 0.02, genes with the largest differences in production rates between “missing” and “no missing” data were MSN2, AFT2, HAP4, ASH1, and CIN5. At alpha 0.002, only CIN5 and ASH1 were still in the top five genes with the largest differences, and as noted before, the magnitude differences were small.

With regard to the threshold b values (Table 7), when the alpha value was 0.002, the mean difference between “missing” and “no missing” estimations was 0.5553 and the standard deviation was 1.1317, as compared to when the alpha value was 0.02, where the mean difference was 0.0177 and the standard deviation was 0.0249. All 16 genes had reduced differences in b when alpha was 0.002, except for ABF1 and ZAP1 whose b values remained 0 because they had no incoming edges. Again, the range of b values was reduced when alpha was changed from 0.002 to 0.02. The genes with the largest magnitude differences at alpha 0.02 were AFT2, CIN5, HMO1, and MSN2, in descending order. These same four genes had the largest magnitude differences in b at alpha 0.002, albeit in a slightly different order.

Considering the estimation of all three parameter values (P , b , and w) together, for alpha 0.002, the genes that had large differences in all parameters between the “missing” and “no missing” data conditions were MSN2, AFT2, and CIN5. When the alpha value was 0.002, the two edge weights with the largest difference were regulated by MSN2, which also had the largest difference for production rates and had the fourth largest difference in b values. AFT2 had the second largest difference for P and the largest for b . Looking at the GRN structure, AFT2 is not a regulator and is only regulated by MSN2. Additionally, the difference in b value is more than four times larger than the next highest difference. Similarly, the MSN2 to AFT2 weight had a

much larger difference than the other weights when the alpha was 0.002. With an alpha of 0.02, both of these differences were reduced. At alpha 0.02, only CIN5 had large differences in all three P , b , and w parameters, although MSN2 had large changes in b and two associated weights, and AFT1 had a large difference in b . Overall, DB1 seemed much less sensitive to the “missing” versus “no missing” expression data condition than DB5, although like with DB5, MSN2 had the greatest parameter variation across conditions.

However, for DB1, many more differences in weight parameter values were seen when looking at the differences between the two alpha values comparing “missing” to “missing” and “no missing” to “no missing” expression data conditions (Table 5). Seven weights changed sign. The four largest magnitude differences were for MSN2 to HAP4, MSN2 to CIN5, MSN2 to YOX1, and HMO1 to HAP4. Note that MSN2, HAP4, and YOX1 showed large differences in all three parameter values (P , b , and w) for the different model conditions for DB5. So in the case of DB1, while the GRN seemed less sensitive to the “missing” versus “no missing” expression data conditions, it was highly sensitive to alpha 0.02 versus 0.002.

We next examine parameter estimation results for DB2 (Appendix A Figure 3, Tables 8, 9, and 10) under the four conditions of “no missing” expression values, “missing” values at alpha 0.002 and 0.02. As can be seen in Appendix A Figure 3 and Table 8, when the alpha value was 0.002, there were three estimated weights that changed sign between the “missing” and “no missing” expression value input workbooks. However, when the alpha value was increased to 0.02, there were no weights that changed signed. Comparing results for “no missing” data between the different alpha values, there were two weights that changed sign and only one that changed sign comparing the results for the “missing values” between the different alphas. When the alpha value is smaller, the range of weight estimations is larger as seen in Table 8 and by the

thicker arrows in Appendix A Figure 3. For example, for “missing” data at alpha 0.002, weights range from -4.32 to 3.197 (7.517 magnitude difference), for “missing” data at alpha 0.02, weights range from -1.571 to 1.366 (2.937 magnitude difference), with the ranges for the “no missing” data conditions being extremely similar. When the alpha value is decreased, the magnitudes of the weight values are smaller but also agree more between the “missing” and “no missing” input workbooks.

Table 8. Optimized weight parameters for DB2 from input workbooks where expression values were missing (“missing”) versus input workbooks where the missing values were replaced by the average value for that gene and timepoint (“no missing”) when the alpha value was 0.002 or 0.02. Dark blue cells correspond to the largest magnitude negative weights, and dark red cells correspond to largest magnitude positive weights. White cells correspond to a weight value close to zero. “Absolute difference” is the absolute value of the difference between the weights for “missing” values and “no missing” values for alpha of 0.002 and 0.02. The rank for each alpha was determined by sorting the weights based on absolute difference. A small rank corresponds to a small absolute difference.

Regulator	Target	w when α value = 0.002 (missing)	w when α value = 0.002 (no missing)	w when α value = 0.02 (missing)	w when α value = 0.02 (no missing)	Absolute difference (missing - no missing) α value = 0.002	Absolute difference (missing - no missing) α value = 0.02	Rank α = 0.002	Rank α = 0.02
CIN5	HAP4	0.148	0.065	0.072	0.032	0.082	0.040	16	19
CIN5	SFP1	-0.061	-0.120	-0.062	-0.069	0.059	7.161E-03	13	9
CIN5	STB5	-0.232	-0.213	-0.287	-0.304	0.018	0.017	9	13
CIN5	YHP1	0.182	0.176	0.243	0.242	5.533E-03	1.302E-03	3	6
GCR2	MSN2	-3.399	-3.171	-2.391	-2.286	0.228	0.105	20	22
GLN3	MGA2	0.056	0.056	0.037	0.038	1.973E-04	2.399E-04	1	1
HMO1	CIN5	0.780	0.786	0.664	0.682	0.006	0.018	2	14
HMO1	HAP4	-0.661	-0.568	-0.299	-0.277	0.093	0.023	17	16
HMO1	HMO1	0.744	0.734	0.878	0.877	0.010	2.567E-04	6	2
HMO1	MSN2	0.212	0.195	0.196	0.197	0.016	8.969E-04	8	4
HMO1	YOX1	0.617	0.581	0.540	0.531	0.037	9.082E-03	12	10
MSN2	CIN5	-2.228	-2.292	-1.399	-1.334	0.064	0.065	14	20
MSN2	HAP4	3.197	3.131	1.366	1.367	0.066	1.156E-03	15	5
MSN2	SFP1	1.209	-0.046	0.888	0.887	1.254	7.431E-04	24	3
MSN2	SWI4	-0.791	-0.781	-1.113	-1.093	0.010	0.020	5	15
MSN2	YHP1	-0.898	-0.922	-0.319	-0.233	0.024	0.086	11	21
MSN2	YOX1	-2.086	-1.962	-1.384	-1.273	0.124	0.111	19	23
SFP1	SWI5	0.766	-1.654	0.289	0.304	2.420	0.015	25	12
STB5	HAP4	-0.495	-0.884	-0.904	-1.020	0.389	0.116	22	24
STB5	SFP1	-0.316	-0.859	-0.425	-0.430	0.543	0.005	23	8
SWI4	HAP4	-2.764	-2.466	-1.571	-1.425	0.299	0.146	21	25
SWI4	YHP1	-4.320	-4.331	-0.972	-0.995	0.011	0.023	7	17
SWI4	YOX1	-0.105	0.007	0.336	0.310	0.112	0.026	18	18
YHP1	GLN3	0.576	0.596	0.233	0.246	0.020	0.013	10	11
YHP1	RDS3	0.691	0.699	0.536	0.530	8.183E-03	5.410E-03	4	7

When the alpha value was 0.002, the highest magnitude weight difference was 2.42 for SFP1 to SWI5 and the smallest difference was 1.973×10^{-4} for GLN3 to MGA2 (Table 8). The average difference was 0.236 and the standard deviation was 0.527. When we increased the alpha to 0.02, the highest magnitude difference was only 0.146 for SWI4 to HAP4 with an average of

0.0342 and a standard deviation of 0.0435. With alpha values, we can see that similar to DB1, the differences are much less extreme than for DB5.

For the production rates (Table 9), when the alpha value was 0.002, the mean difference between the “missing” and “no missing” data condition was 0.1543 and the standard deviation was 0.2546, as compared to when the alpha value was 0.02, the mean difference was 0.0154 and the standard deviation was 0.0235. Again, the range of values was reduced when changing alpha from 0.002 to 0.02. For the threshold b values, when the alpha value was 0.002, the mean difference between the “missing” and “no missing” data conditions was 0.350, the standard deviation was 0.571, and the sum of the differences was 4.89. When the alpha was 0.02, the mean difference was 0.03, the standard deviation was 0.03, and the sum of the differences was 0.42. Thus, we see that when the alpha was increased, most of the major discrepancies between b values were resolved.

We can still look at specific weights to draw connections between estimating weights, production rates, and threshold b values. For alpha 0.002, the largest difference in estimated weights between “missing” and “no missing” data was SFP1 to SWI5 followed by MSN2 to SFP1 and STB5 to SFP1 (Table 8). As we see in Tables 9 and 10, SFP1 has the second largest difference in production rates and the largest difference for threshold b value. MSN2, SWI5, and STB5 are also in the top five for largest differences in both P and b for alpha 0.002. When alpha was increased to 0.02, only MSN2 had a relatively large difference in P between “missing” and “no missing” data at 0.0801 and b at 0.0916. The difference in the SFP1 to SWI5 edge weight for “missing” versus “no missing” data dropped in rank from the largest to 12th smallest, the weight for MSN2 to SFP1 dropped from second largest to third smallest, and STB5 to SFP1 dropped from third largest to 8th smallest. When differences in P and b values resolved through increasing

the alpha, the differences between the estimated weights also became smaller. Overall, the estimated parameters for DB1 were far less sensitive to changes in model conditions than DB2 (more sensitive) or DB5 (most sensitive), although MSN2 stands out as problematic in all three GRNs.

Table 9. Optimized production rates for DB2 from input workbooks where expression values were missing (“missing”) versus input workbooks where the missing values were replaced by the average value for that gene and timepoint (“no missing”) when the alpha value was 0.002 as compared to an alpha value of 0.02. “Absolute difference” is the absolute value of the difference between the production rates for “missing” values and “no missing” values for alpha of 0.002 and 0.02. Cells with darker green indicate a larger difference in magnitude and lighter green indicates a smaller difference in magnitude.

Gene ID	P when α value = 0.002 (missing)	P when α value = 0.002 (no missing)	P when α value = 0.02 (missing)	P when α value = 0.02 (no missing)	Absolute difference (missing - no missing) α value = 0.002	Absolute difference (missing - no missing) α value = 0.02
CIN5	1.4313	1.5957	0.9758	0.9419	0.1644	0.0339
GCR2	0.2256	0.2251	0.2443	0.2428	0.0005	0.0015
GLN3	0.4183	0.4176	0.3431	0.3421	0.0007	0.001
HAP4	1.1027	1.0784	0.6951	0.6981	0.0243	0.003
HMO1	0.4847	0.5112	0.4026	0.4023	0.0265	0.0003
MGA2	0.2725	0.2725	0.2733	0.2733	0	0
MSN2	2.3112	2.1533	1.6909	1.6108	0.1579	0.0801
RDS3	0.2968	0.2958	0.2122	0.2119	0.001	0.0003
SFP1	0.9295	1.4091	0.9159	0.9252	0.4796	0.0093
STB5	0.3408	0.6061	0.2497	0.2579	0.2653	0.0082
SWI4	0.3093	0.3353	0.6670	0.6576	0.026	0.0094
SWI5	0.2531	1.1526	0.2216	0.2212	0.8995	0.0004
YHP1	1.8748	1.8777	0.4012	0.3798	0.0029	0.0214
YOX1	1.4887	1.3766	0.8692	0.8224	0.1121	0.0468

Table 10. Optimized threshold b values for DB2 from input workbooks where expression values were missing (“missing”) versus input workbooks where the missing values were replaced by the average value for that gene and timepoint (“no missing”) when the alpha value was 0.002 as compared to an alpha value of 0.02. “Absolute difference” is the absolute value of the difference between the production rates for “missing” values and “no missing” values for alpha of 0.002 and 0.02. Cells with darker green indicate a larger difference in magnitude and lighter green indicates a smaller difference in magnitude.

Gene ID	b when α value = 0.002 (missing)	b when α value = 0.002 (no missing)	b when α value = 0.02 (missing)	b when α value = 0.02 (no missing)	Absolute difference (missing - no missing) α value = 0.002	Absolute difference (missing - no missing) α value = 0.02
CIN5	-0.3288	-0.2621	0.1110	0.1821	0.0667	0.0711
GCR2	0	0	0	0	0	0
GLN3	0.7138	0.7298	-0.0112	-0.0057	0.016	0.0055
HAP4	2.5252	2.1602	0.4400	0.3583	0.365	0.0817
HMO1	1.6632	1.7363	1.5375	1.5348	0.0731	0.0027
MGA2	-0.0249	-0.0250	-0.0359	-0.0355	0.0001	0.0004
MSN2	-1.5375	-1.3750	-0.9503	-0.8587	0.1625	0.0916
RDS3	1.0639	1.0661	0.2644	0.2565	0.0022	0.0079
SFP1	0.7987	-0.8477	0.2071	0.1805	1.6464	0.0266
STB5	0.7486	1.4107	0.1835	0.1179	0.6621	0.0656
SWI4	-0.6971	-0.5403	-0.1214	-0.0975	0.1568	0.0239
SWI5	1.0776	-0.5321	0.0405	0.0494	1.6097	0.0089
YHP1	-1.7663	-1.8025	0.3806	0.4027	0.0362	0.0221
YOX1	-0.3376	-0.2351	0.2234	0.2377	0.1025	0.0143

After performing this analysis, we wondered if we could explain these differences in parameter sensitivity by looking at the optimization diagnostics and network structure similarities and differences between the three GRNs. Table 11 shows the penalty term, LSE, minLSE, LSE:minLSE ratio, and iteration count for the models discussed in Appendix A Figures 1-3 and Tables 2-10. Some patterns emerge. For each GRN, the penalty terms are larger at alpha 0.002 versus 0.02, which is expected from our observation that there was a smaller range of estimated w , P , and b values for all networks at alpha 0.02. In almost all cases, the penalty term is also smaller for each network when comparing the conditions of “missing” versus “no missing” values. The iteration counts are also larger at alpha 0.002 versus 0.02, which is also expected because of the overfitting occurring with a smaller alpha (also see Figure 2). For each

pair of models, the minLSE is smaller in the “no missing” values condition versus the “missing” values is expected because replacing missing values with the average value for that gene and timepoint would reduce the variation in the replicates. When examining the LSE:minLSE ratios, DB2 has the four smallest values for the four model conditions, followed by DB1 and then DB5. However, the four ratios for DB1 are closer to DB5 than either of them is to DB2. It should be noted that the minLSE values for DB2 and DB5 are similar, so the better LSE:minLSE ratios for DB2 are determined by smaller LSE values in the numerator. In other words, the LSE:minLSE error is larger for DB5 than for DB1, which has larger minLSEs, because DB5’s LSE is correspondingly larger. While the range of values for the LSE:minLSE ratio is quite small (1.344 to 1.448), the ranking of the ratios seems to track with our analysis of the parameter estimation results for the three GRNs.

Table 11. Optimization diagnostics for DB1, DB2, and DB5 for parameter estimations performed with input workbooks at alpha 0.002 and 0.02 and with missing expression datapoints in the input workbooks (“missing values”) or the missing datapoints replaced by the average value for that gene, strain and timepoint (“no missing values”). Penalty term is the vector of parameters. LSE is least squares error between model simulated and experimental expression data; minLSE is the minimum LSE possible given the variation in the input expression data. Iteration count is the number of number of times the ODE was solved before the LSE was minimized.

Parameter	Iteration Count	Penalty Term	LSE	minLSE	LSE:minLSE Ratio	Rank
DB1 (α value = 0.002) with missing values	84804	2.593	0.819	0.577	1.421	8
DB1 (α value = 0.002) with no missing values	74525	1.891	0.785	0.562	1.397	5
DB1 (α value = 0.02) with missing values	43787	0.598	0.825	0.577	1.431	11
DB1 (α value = 0.02) with no missing values	43919	0.615	0.787	0.562	1.401	6
DB2 (α value = 0.002) with missing values	63902	1.916	0.671	0.499	1.344	1
DB2 (α value = 0.002) with no missing values	53846	1.893	0.663	0.488	1.358	2
DB2 (α value = 0.02) with missing values	22177	0.6	0.678	0.499	1.358	2
DB2 (α value = 0.02) with no missing values	25836	0.564	0.67	0.488	1.371	4
DB5 (α value = 0.002) with missing values	111242	2.507	0.7	0.497	1.408	7
DB5 (α value = 0.002) with no missing values	61273	2.34	0.692	0.485	1.427	9
DB5 (α value = 0.02) with missing values	30248	0.967	0.711	0.497	1.430	10
DB5 (α value = 0.02) with no missing values	27897	0.889	0.702	0.485	1.448	12

The take-home messages from the analysis of DB1, DB2, and DB5 are as follows:

- The range of estimated parameter values is larger when alpha is smaller. This, in turn, influences the magnitude of the differences seen when estimating parameters with “missing” versus “no missing” data.
- DB5 is most sensitive to changes in the input workbooks with regard to both “missing” and “no missing” expression data and alpha values. DB1 is less sensitive to “missing” data but is still sensitive to changes in alpha. On the other hand, DB2 parameter estimation is robust to differences in alpha or input expression data.
- The LSE:minLSE ratios track with the above observations, even though the changes in magnitude of the ratios are small.
- Across the three networks, parameters associated with certain genes were more sensitive to conditions than those of other genes. MSN2 generally had the most discrepancies across the board.
- Because DB2 did not have *ZAP1* in its network, it did not include the *Azap1* expression data as input. This circumstance cannot be ruled out as a driver for the robustness of the estimation relative to the other two GRNs.

Even with small networks of 14-16 genes, the GRN structure is complex and difficult to analyze. DB1, DB2, and DB5 share nine genes, and six genes are unique to a single network. DB1 has four unique genes and DB2 only has two genes unique to its network, MGA2 and RDS3, which were not implicated in large parameter differences. All of the genes in DB5 are shared with at least one other network. We hypothesize that DB5 has a more densely connected network structure than the other two networks which could cause estimation to be more difficult because changes in downstream parameter values can have a larger impact on the estimation of

each weight. Because of these variations and these questions, a sensitivity assay in an even smaller system was designed to better understand why this might be occurring.

3-gene, 4-edge “toy” network sensitivity assay

In designing the sensitivity assay, we wanted to explore an even smaller network to reduce the number of parameters the model was estimating. For each input workbook when production rates, threshold b values, and weights are all estimated, the number of parameters is twice the number of genes plus the number of edges. Thus, for a small 3-gene, 4-edge network, there were ten parameters being estimated. As explained in the Methods, to complete this assay, we generated simulated gene expression data from known parameters and then estimated the parameters using the simulated gene expression data. From the output, we then compared the known input weight parameters to the estimated output weight parameters and calculated the LSE to determine the difference between the expected and observed output.

Since we were creating these input workbooks with simulated expression data, they inherently had no noise. Thus, we needed to adjust the optimization parameters to reflect the noiseless data. We adjusted the alpha value, tolfun, and tolX. When optimizing parameters with an iterative approach, there are two main ways the model knows when to stop iterating; when the two successive solutions are close to each other (tolX) or when the function it is trying to minimize gets close to zero (tolfun). The model stops iterating when one of the above conditions is met. We determined that an alpha value of 2×10^{-8} and a tolfun and tolX value of 2×10^{-8} were sufficient. In comparison, DB5 had an alpha value of 0.02 and tolfun and tolX values of 1×10^{-6} (Table 12).

Table 12. Optimization parameters of DB1, DB2, DB5 compared to optimization parameters of the 3-gene, 4 edge networks.

Parameter	DB1, DB2, and DB5	3-gene, 4-edge networks
alpha	2.00E-02 or 2.00E-03	2.00E-08
kk_max	1	1
MaxIter	1.00E+08	1.00E+08
TolFun	1.00E-06	1.00E-08
MaxFunEval	1.00E+08	1.00E+08
TolX	1.00E-06	1.00E-08

Later, we realized that an L-curve should have been performed to determine the best alpha value for the new smaller network. Thus, although an alpha value had already been selected, a post hoc L-curve was run to determine if this was a good choice (Figure 3).

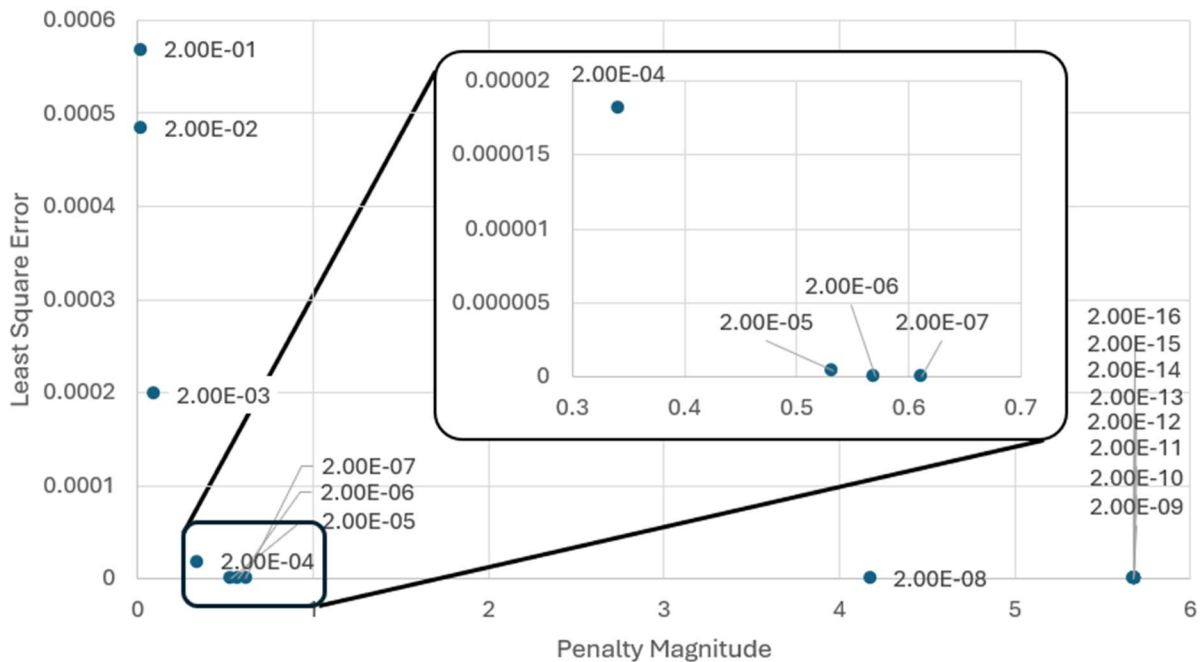


Figure 3. L-curve for the bimutual 3-gene, 4-edge network to determine the alpha value. Values next to the datapoints indicate what alpha value the point corresponds to. Inset is zoomed in to show the elbow of the curve more clearly.

Looking at the L-curve, it seems an alpha of between 2×10^{-4} and 2×10^{-5} would have been right in the elbow of the curve, and the choice of 2×10^{-8} might have been too small of an alpha.

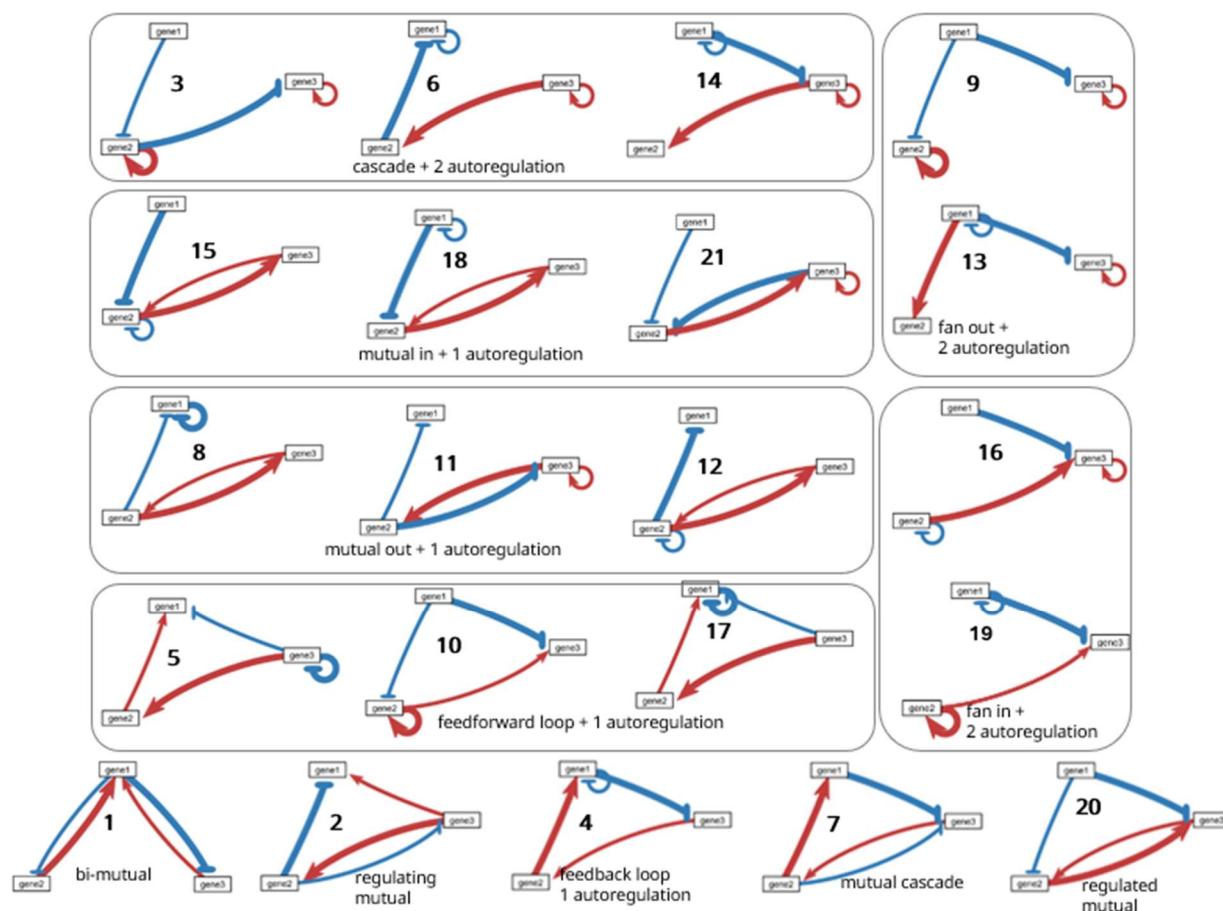


Figure 4. All 21 possible 3-gene, 4-edge networks with the type of motif labeled and grouped by type. Blue, blunt arrows indicate repression and red, pointed arrows indicate activation with thicker arrows representing the larger magnitude weight (-2, 2) and thinner arrows representing the smaller magnitude weight (-1, 1). The number of each network corresponds to the rank in Table 13.

Input workbooks with these optimization parameters were then run with all 21 different 3-gene, 4-edge network motifs in an effort to determine the influence of network structure upon model results. As mentioned earlier, each edge was arbitrarily assigned a weight value of -2, -1, 1, or 2 which can be seen by the thickness and color of the arrows in Figure 4.

Table 13. LSE between true weights and estimated weights for all 3-gene, 4-edge networks when estimating each combination of parameters. Each network was ranked based on the sum of LSE for the four conditions. **Bold** indicates the largest LSE per network, and *italics* indicates the smallest LSE per network.

Rank	Estimate P, b, w	Estimate P and w b held constant	Estimate b and w P held constant	Estimate w P and b held constant	Sum
1	0.0209	1.0243E-07	3.6031E-08	<i>1.6927E-12</i>	0.0209
2	0.0653	1.3301E-06	9.2078E-09	<i>6.2412E-11</i>	0.0653
3	0.6416	0.1982	1.7491E-09	<i>1.2760E-09</i>	0.8397
4	1.5532	0.0002	<i>3.4341E-08</i>	8.4351E-09	1.5535
5	2.5158	0.7600	2.8491E-07	<i>4.2708E-10</i>	3.2758
6	0.0214	3.7478	7.3398E-06	<i>1.3415E-11</i>	3.7692
7	2.2486	1.9650	8.1226E-10	<i>3.9196E-11</i>	4.2136
8	4.1162	0.2369	3.6304E-06	<i>4.3225E-11</i>	4.3531
9	3.8276	0.0231	1.1926	<i>3.4098E-09</i>	5.0433
10	1.1138	5.9429	1.1992E-09	<i>4.9337E-10</i>	7.0567
11	0.0798	9.1072	8.6779E-11	<i>8.4445E-12</i>	9.1870
12	4.5806	7.7364	3.0405E-06	<i>2.3978E-08</i>	12.3170
13	4.4742	<i>2.6741E-07</i>	4.5016	4.4999	13.4757
14	6.8500	<i>3.7518E-09</i>	4.4999	4.5001	15.8499
15	5.9286	<i>2.9181</i>	4.3002	3.7478	16.8947
16	8.8410	<i>0.0004</i>	7.1053	7.4670	23.4137
17	5.2996	25.0259	0.0053	<i>0.0032</i>	30.3341
18	0.6019	32.9598	4.5124E-10	<i>2.6854E-10</i>	33.5616
19	0.2739	33.3092	1.0115E-09	<i>6.4618E-10</i>	33.5831
20	26.9602	22.4204	<i>9.9182</i>	9.9871	69.2858
21	4.9845	172.8924	2.5940E-09	<i>7.9428E-10</i>	177.8768

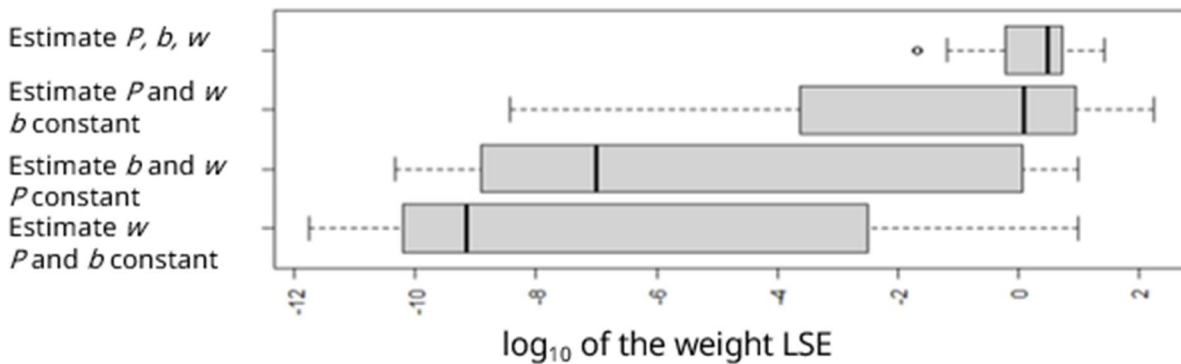


Figure 5. Log₁₀ of the weight LSE (Table 13) for the 21 network motifs (Figure 4) when weights were arbitrarily chosen and the model was run with each combination of parameters being estimated.

In addition to determining the influence of network structure, we also wanted to determine whether the model is sensitive to the number of parameters being estimated. We did this through running GRNmap four times for each input workbook; first estimating production rates, weights, and threshold b values, second estimating production rates, weights, and holding threshold b values constant, third estimating weights and threshold b values, and holding production rates constant, and fourth estimating weights and holding production rates and threshold b values constant. Table 13 lists the weight LSEs for each of the 21 networks run under these four conditions, along with the sum of the LSEs for each network, which was used to rank the network models from best to worst (see also numbering in Figure 4). Thirteen of the 21 networks had their largest LSE value when all three parameters were estimated, and 7 networks had their largest LSE value when b was held constant and P and w were estimated (Table 13, **bold**). One network (ranked 13th) had its largest LSE value when P was held constant and b and w were estimated, but this value was within a tenth of the values that occurred when either all three parameters were estimated or when w only was estimated. Fifteen of the 21 networks had their smallest LSE value when only w was estimated (Table 13, *italics*). Four networks had their smallest LSE value when P and w were estimated with b held constant, and two had the smallest when b and w were estimated and P was held constant. For the nine top-ranked networks, a pattern generally holds where the LSE is smallest when only w is estimated, next smallest when b and w are estimated, next smallest when P and w are estimated, and largest when all three parameters are estimated. However, below these top nine networks, the pattern falls apart. Four of the five bottom-ranked networks have the largest LSE value when w and P are estimated, and would have ranked higher except for that outlier. Figure 5 is a box and whiskers plot of the $\log_{10}$ of the weight LSEs under the four conditions, and shows that the spread of the data is quite large

under all four conditions, spanning several orders of magnitude. The median LSE decreases when only b 's and weights are estimated, and even more when only weights are estimated. In other words, the median LSE is largest when P is estimated, suggesting that it is difficult for GRNsight to perform weight and b parameter estimation when production rates are also being estimated. As seen in Figure 5, while the median is the largest when all parameters are estimated, the spread of the data is not as large as under the other conditions where P and/or b are held constant. It is possible that when the model has three parameters to adjust, it can achieve a good fit to the expression data, while sacrificing parameter accuracy. Additionally, since the weights for these networks were assigned arbitrarily, it could be the case that for a few of the network structures, the assigned weights do not make much biological sense, discussed later.

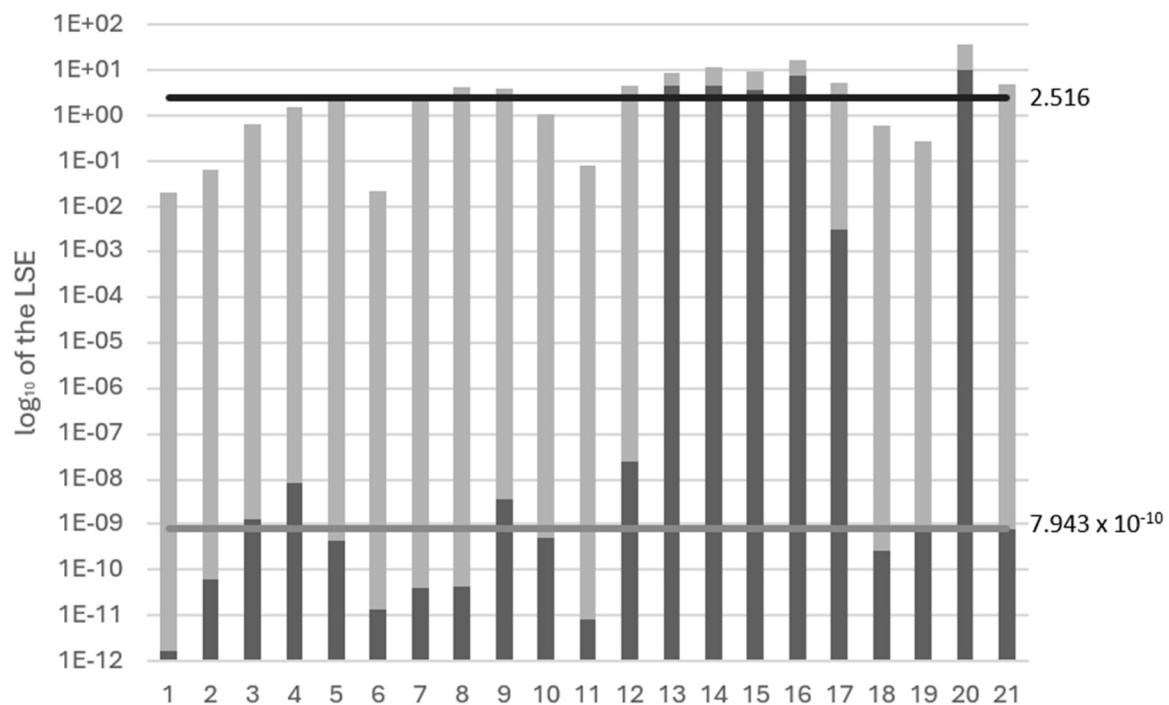


Figure 6. $\log_{10}$ of the weight LSE of 21, 3-gene, 4-edge networks numbered how they are ranked in Fig. 4 and Table 13. Light gray represents LSE when all parameters are estimated and dark grey represents when only w is estimated. X axis is the rank from Table 13. Horizontal lines represent the median value of all of the LSEs.

Figure 6 shows a stacked bar chart of the weight LSEs when just w was estimated and when all parameters were estimated. We can see that ranking based on the sum of the LSE from all four conditions is different than the ranking would be based on a single LSE value. In particular, networks ranked 13, 14, 15, 16, and 20 stand out as having large LSE values when only w is estimated with the same order of magnitude as when all three parameters are estimated. From Table 13 we can see that the smallest LSE for networks 13, 14, 15, and 16 is when b is held constant. Network 15 is the same network type as networks 18 and 21, mutual in with one autoregulatory edge. These three networks exhibit interesting behavior with respect to their LSEs across conditions. Network 21 stands out as an obvious outlier because it has the largest LSE for all the networks and conditions (172.8924, Table 13) when weights and production rates are estimated and b is held constant. In all other conditions, it has much smaller LSE values, and it would have been ranked among the top nine networks. Network 18 follows the same pattern of LSEs as network 21, but its largest LSE when w and P are estimated is not quite as large as network 21 (32.9598, Table 13). Network 18 is actually a more extreme case than network 21 with the largest LSE being about 55 times bigger than the second largest LSE. For network 21 the largest LSE is about 35 times larger than the next largest LSE. Interestingly, network 15 has the opposite pattern where it seems to be following the trend of a decreasing LSE moving to the right in the table, but its smallest LSE is when b is held constant, although all LSEs are close to each other in value. Looking at the network structures themselves (Figure 4), we see that for network 15, gene 2 has three incoming weights, two of which are repressors and one is an activator. Gene 3 has the third incoming weight, which is an activator. Thus, for this network structure, the edges are primarily regulating one target gene. In contrast, for network 18, gene 1 has one incoming weight, gene 2 has two incoming weights, and gene 3 has one incoming

weight. In network 21, gene 2 has two incoming weights and gene 3 has two incoming weights. In this case it is possible that when restricting the number of parameters estimated when the weights are more spread out, it might be more difficult for GRNmap to find the weights accurately.

In summary, we see that the median LSE when production rates and threshold b 's were held constant was relatively similar to when only production rates were held constant (Figure 5). Additionally, the median LSE is much higher when production rates are estimated and similar when all three parameters are estimated and when b is held constant. Thus, it seems likely it is estimating this specific parameter, P , not the overall number of parameters estimated that has the greatest impact on model accuracy.

While we were hoping that this data would give more insight into the effect that network structure has on parameter estimation, we soon realized that for each network structure, there is a total of 24 networks that could be built by permuting the assigned weight values $(-2, -1, 1, 2)$ between the four edges. Thus, these 21 original networks serve as representatives for the network type (Figure 4), but to investigate network structure itself, all permutations need to be modeled to ensure that the arbitrarily assigned weights did not overly influence the performance of the network structure relative to the other networks.

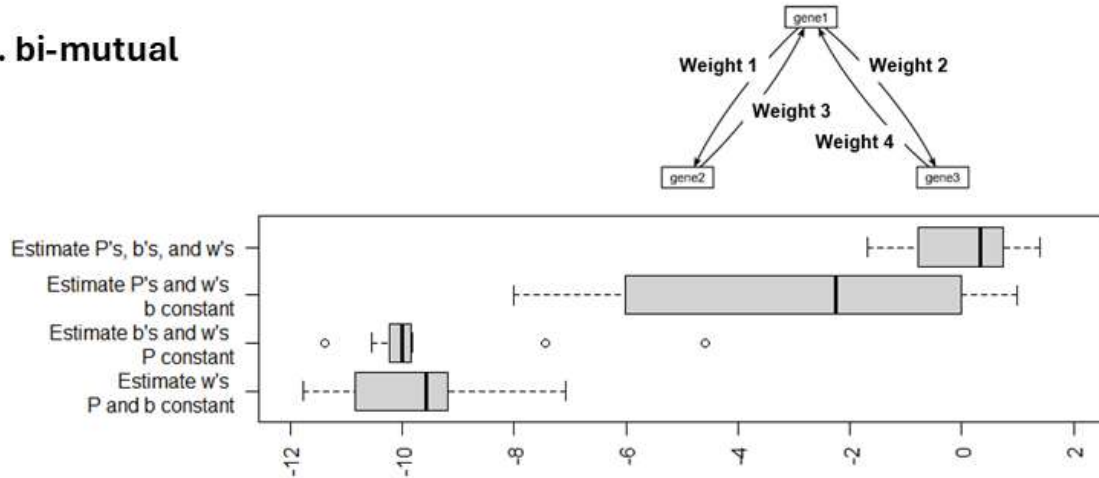
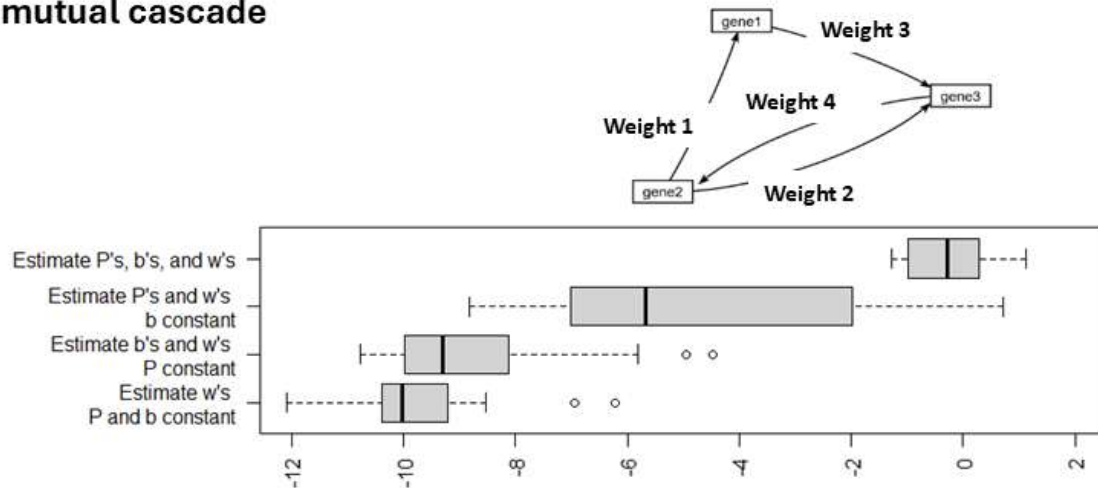
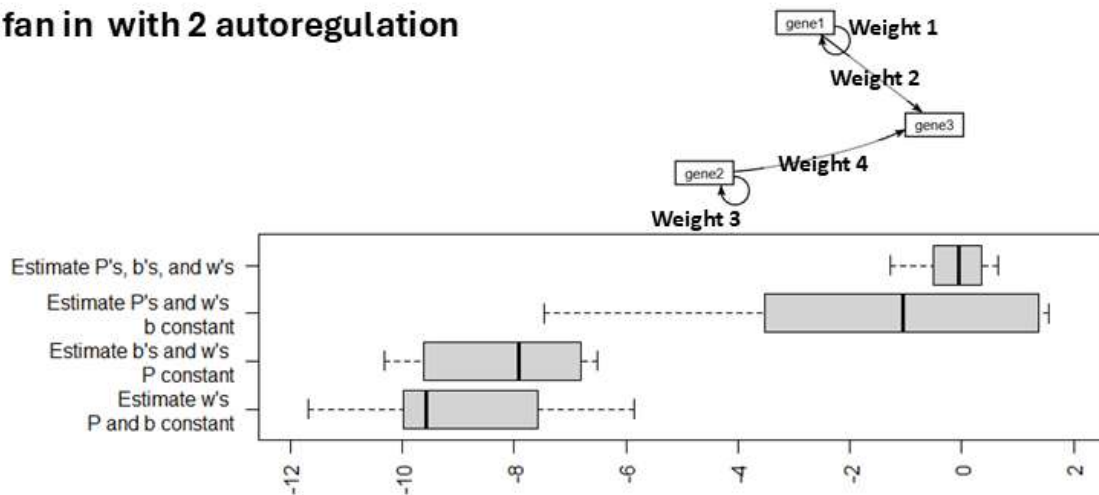
A. bi-mutual**B. mutual cascade****C. fan in with 2 autoregulation**

Figure 7. $\log_{10}$ of the weight LSE for the 24 weight assignment perturbations for the networks originally ranked 1st (A), 7th (B), and 19th (C) for all different combinations of parameters held constant.

To see if and how the arbitrary weight choice influenced network rank, we chose three different networks to permute: the network 1 (bi-mutual), network 7 (mutual cascade), and network 19 (fan in with two autoregulatory edges). Figure 7 shows box and whiskers plots of the $\log_{10}$ of the weight LSEs for the three networks when the assigned weights were permuted, under the four conditions. As was seen in Figure 5 with the original 21 networks, the spread of the data is quite large under all four conditions, spanning several orders of magnitude. When the threshold b values are held constant the resulting LSEs have the largest range over the four conditions. Overall, the network originally ranked 7th generally has a smaller distribution for most conditions compared to the other networks, and the four medians are better than for the other two permuted networks. It noticeably performs better than the other two in terms of median and range when b is held constant. Thus, if we take all of the permutations into account, this mutual cascade network performs better than the other two.

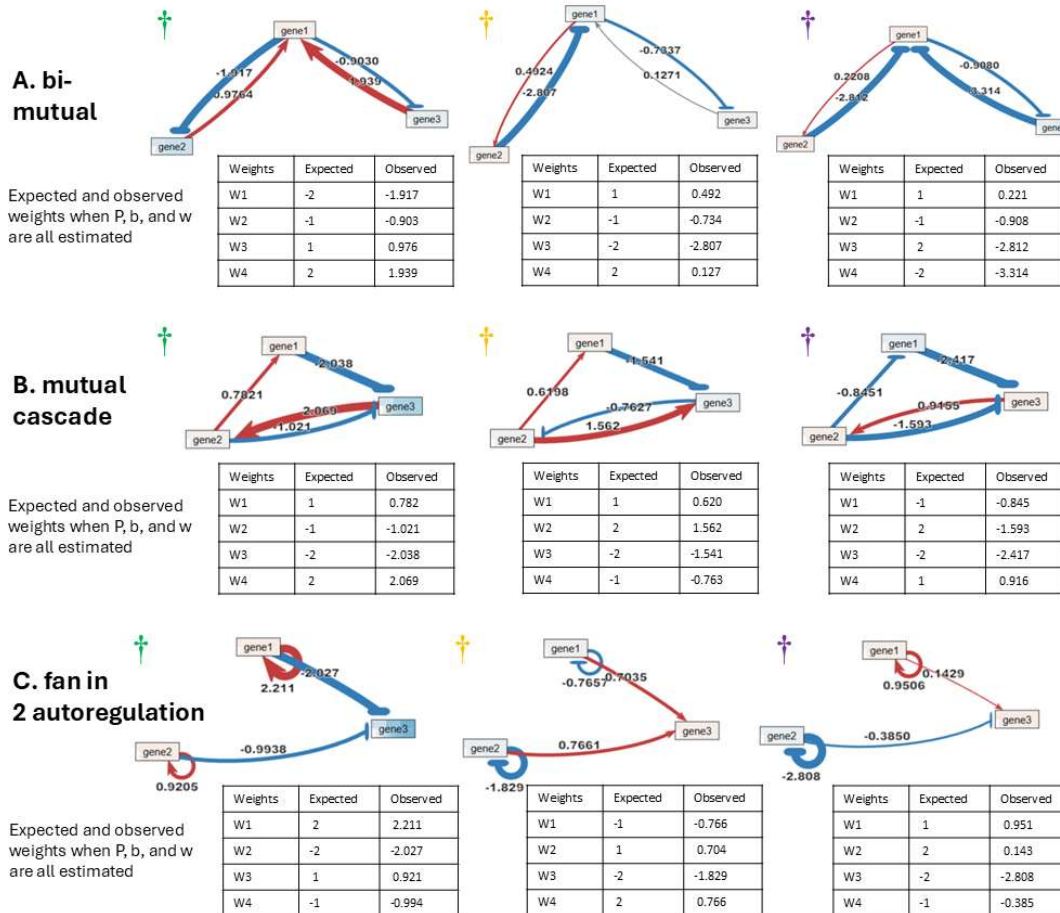


Figure 8. Weight estimation for selected permutations of networks ranked 1st (A), 7th (B), and 19th (C) when all parameters are estimated. The † corresponds to Appendix 1 Figure 4. The green dagger corresponds to the network that was ranked best, the orange dagger represents a network in the middle, and the purple dagger represents the network ranked last.

Appendix A Figure 4 shows a stacked bar chart of the LSE values when only w is estimated and when P , b , and w are estimated, for each network structure. Overall, the medians of the weight LSE when only w was estimated and when P , b , and w were all estimated are within one order of magnitude of each other, with network 7 having the smallest LSEs of the three networks under both conditions. Appendix A Figure 4 shows that the network weights that were originally assigned in the first 21 networks do have an effect on the LSE. For example, in Appendix A Fig. 4a we see that the network with the weights that were originally assigned has

the second lowest LSE of all 24 networks compared to the network in Appendix A Figure 4b where the originally assigned weights resulted in a ranking of 19 and in Appendix A Figure 4c a ranking of 22. Thus, the original ranking of the 21 networks does not give us any insight into whether some networks performed “better” or “worse” than others because of their structure, but rather we can see that the assigned weight values in addition to structure impact the accuracy of weight estimation. Additionally, Fig. 8 allows us to visualize the effect of some of the weight permutations when P , b , and w were all estimated. We see that the best performing networks have weight values that are generally close to the expected weights, but the worst performing networks are not accurate with two of the networks even having a weight that changed sign.

Appendix A Fig. 4a reveals a pattern in the LSEs with respect to the weight assignments for the bi-mutual network. The LSEs are quite similar in pairs of networks when the weight values for weight 1 and weight 2, and weight 3 and weight 4 are switched. Moreover, these pairings of two networks are paired with two other networks where the two negative weight values (-2, -1) are swapped with each other and the two positive weight values (1, 2) are swapped with each other (e.g., see the four left-most bars in Appendix A Figure 4a). This pattern makes sense because the bi-mutual network structure is symmetrical. However, for the other two networks (Appendix A Figure 4b, c) there does not seem to be a clear pattern in how weight assignment affects parameter estimation. The network in Appendix A Fig. 4c was specifically selected because we were interested to see if it was more difficult for our model to accurately estimate weights for networks with autoregulation. I had originally hypothesized that GRNmap did not handle autoregulation very well based on some of the results from the initial ranking. However, comparing the LSE distribution in Fig. 7c to that of the other two networks, it does not seem like the estimation of the network 19 permutations are less accurate than networks 1 or 7,

despite the fact that this network had two autoregulatory loops. It seems that the weight estimation is generally more accurate for networks with positive autoregulation than negative autoregulation. When comparing the sum of the expected weights for this network, I compared the permutations ranked first through 12th and the permutations ranked 12th through 24th, and the sum for the first set for weights 1 and 3 was positive—so there was more positive autoregulation.

Ultimately, the data shown throughout the results leads us to have two major conclusions. First, estimating production rates from limited gene expression data is difficult for GRNmap to do. Secondly, performance of the model is influenced by both the network structure and the edge weights, and determining the relative influence that each has alone will require further investigation.

Discussion

At the onset of this project, we aimed to answer the question of why input workbooks with missing expression values have different parameter outputs than those where the missing values were replaced by the average value for that gene, strain, and timepoint. Past members of the Dahlquist lab had designed seven different candidate gene regulatory networks while trying to answer the questions: which regulatory transcription factors are involved in the budding yeast cold shock response and what are their relative levels of influence. Of these seven networks, DB5 was investigated the most. Because more analysis was performed on this network, we started working with this network first to investigate the impact of the missing gene expression values on model output because of previously noted discrepancies. In an effort to minimize these differences we changed the alpha value from 0.002 to 0.02, and the resulting output was still inconsistent. It is notable that this was the first network investigated because it also had

dramatically different results than the other two networks. If we had selected DB1 or DB2 to investigate first, the differences in the estimated parameters would not have been as striking. Thus, the cumulative results of our analysis of these three networks turned our focus away from why missing input expression values impacted parameter estimates, but rather to what was unique about DB5 that caused it to have such different results than the other two networks.

In comparing DB1, DB2, and DB5, we see that DB5 does not have any genes that are not found in the other two networks, whereas DB1 and DB2 do have some unique genes. It is also of interest to note that the DB2 input workbook is missing the gene expression worksheet for the *Δzap1* strain because this gene is not present in the network. Overall, we see that missing expression values are only a small percentage of the total expression data, but we also see that a large number of these missing datapoints are associated with STB5 and ZAP1 (Table 1). Even though these genes have the most missing expression datapoints, they were not the genes for which most of the discrepancies were found. Generally, more problems arose for MSN2 across all three GRNs which had far fewer missing values. Additionally, ASH1 (which is only in DB5 and DB1) and HAP4 have no missing values at all and were problematic as well. Thus, it seems that the issues are not inherently dependent on which genes have missing data, but on the reduction in variation in the replicate expression data when those values are replaced by their averages for that gene, strain, and timepoint. Because of the unique results from DB5 it seems that the overall impact of the missing datapoints is most dependent on network structure.

We hypothesize that the stability of a graph is somehow related to the network structure, so we can begin to think about that by characterizing some of the aspects of network structure. Since our edges have direction—regulator, target—we have a directed graph or a digraph (Guichard, 2025). While it would be insightful to do an in-depth analysis of each of our network

motifs using graph theory, we will begin by simply focusing on network connectivity by evaluating average degree and density. Average degree can be calculated by dividing the number of edges by the number of vertices, and density can be calculated by $\frac{2(edges)}{(genes)(genes)}$ (Pavlopoulos et al., 2011; Richard, 2024). Thus, the average degrees for DB1, DB2, and DB5 respectively are 2.25, 1.79, and 1.87. Density can be calculated by $\frac{edges}{genes^2}$ (Leclerc, 2008). The densities for DB1, DB2, and DB5 are 0.141, 0.128, and 0.124, respectively. Thus, DB5 has the second highest average degree and is the least dense of the three networks. Looking at the literature, it has been argued that biological networks are likely to be sparsely connected because a sparse graph preserves robustness which is evolutionarily advantageous (Pavlopoulos et al., 2011). It has also been observed that GRNs of *S. cerevisiae*, among other organisms, have connectivity densities lower than 0.1 (Leclerc, 2008). While the densities for our networks were slightly higher than 0.1, since our networks were also much smaller than those considered in Leclerc (2008), it might be the case that we have trimmed down some of the leaf nodes that would make the density of our network lower. While the networks for *S. cerevisiae* in Leclerc (2008) were still only considering transcription factors, they were considering about 50 to 2,000 genes, so that might be where this slight difference comes from.

We can consider one more aspect of graph structure which is the maximal length path. A simple path for digraphs is a finite sequence of nodes where every adjacent node is connected by a directed edge and all edges are pointing down the path (Seymour & Sullivan, 2010). Currently, there is no algorithm for finding the maximal length path, so using GRNsight as a tool for visualizing the network structure, I found the following paths to be of maximal length for each of the networks. For DB1 the path included the following seven genes: ABF1, HMO1, MSN2, CIN5, YHP1, GLN3, GCN4. For DB2, the path included five different genes:

MSN2→CIN5→YHP1→GLN3→MGA2 or HMO1→CIN5→YHP1→GLN3→MGA2. Finally, for DB5 the maximal length path included eight genes:

GCR2→MSN2→CIN5→SFP1→SWI5→ASH1→YHP1→GLN3. DB5 also has two paths with seven genes: HMO1→CIN5→SFP1→SWI5→ASH1→YHP1→GLN3 or

alternately, CIN5→STB5→SFP1→SWI5→ASH1→YHP1→GLN3. Note that in each of these paths there are additional edges which connect to other genes, but these were not included

because either they led to a shorter path or the edge was oriented in the wrong direction. DB2 had the shortest, longest path and was the most robust to changes in model conditions

(missing/no missing data or alpha 0.002/0.02). Graphs with shorter paths are generally more robust because they have multiple routes for regulation rather than relying on a long path for regulating downstream elements (Abdelzaher et al., 2016). Overall, our results show that there is something structurally unique about DB5 which causes it to behave more unstably under different conditions which led us to design a sensitivity assay for 3-gene, 4 edge networks.

In designing this assay, we focused on a much smaller network size so we would be working with a more manageable number of parameters. Our main findings were that estimating production rates from gene expression data is difficult for GRNmap to do, and also that model accuracy is influenced by both network structure and the network weights. While both of these main results are interesting and will help our lab develop better models in the future, it is difficult to make a biological interpretation, especially when we consider that the toy networks were all mathematically possible graphs, rather than just those that are most common in biology. Thus, understanding how network motifs are important and influence larger GRNs is necessary to interpret this work. Some network motifs are more common for different cellular processes, but

overall each network motif carries out specific functions which can be used to understand cellular processes (Alon, 2007).

We can use what is known about network motifs to begin to understand the performance of our network permutations. When beginning to study network motifs, we were interested to see how GRNmap manages autoregulation. Looking at our results, one of the network structures we chose to investigate has two autoregulatory loops. Looking at Appendix A Fig. 4c we see that generally the permutations with positive autoregulation performed better than those with negative autoregulation as seen in the results. Positive autoregulation is been seen to be more accurate when modeling steady state, and since we see that the expression of our genes tend to plateau quite quickly in terms of the modeled expression values, it makes sense that our model would be better able to predict these behaviors (Pinho et al., 2014). Another common network motif is dense overlapping regulons or multi-input motifs (Alon, 2019). While we do not see the fullest extent of this motif with our small networks because each network only has four nodes, this network motif is quite common in yeast when regulating broad biological functions such as stress response (Alon, 2019). Our bimutual motif is similar to a multi-input motif, but it contains reciprocity where the multi-input motif does not. In a 4-gene multi-input motif, two nodes would regulate two edges without the two edges regulating the nodes. The network structure would be more similar to our fan out networks, but with an additional node regulating the same genes that are being regulated. These motifs do appear in our larger networks, DB1, DB2, and DB5. This network motif is quite difficult to study because to fully understand the function of the motif, the input functions in the promotor of each output gene has to be known. Because these functions are not characterized, we cannot fully understand the function of this motif, thus it makes sense for us to break down the motif into smaller more comprehensible units. This is exactly what we are

trying to do in understanding our original networks—DB1 through DB7—and as seen above, we are having these issues which drove us towards small scale analysis. Overall, because our work is based on simulated data, our aim is not to understand the function of any of these networks, but understanding how these motifs behave in biological systems can help us to understand why our model may perform in a certain way (Alon, 2007, 2019).

While the idea of form or structure impacting function is not new to the field of biology, past literature had determined that the structure of a network motif does not strongly determine its function (Ingram et al., 2006). Thus, this would indicate that there is another facet of the network motif that is contributing to its function that is not the structure itself, which then brings to question how we can begin to justify some of the work that we completed. Through redefining what function means in terms of network motif, past literature has found that structural properties of network motifs do determine their functional versatility (Ahnert & Fink, 2016). Previous definitions of function relied more heavily on the results of gene expression patterns, but by defining function at a higher level that is more closely related to a network's topological properties, we can begin to see how form is related. Overall, the work done by Ahnert & Fink (2016) works to justify some of what we have done through investigating small-scale networks. They define function by breaking into functional versatility which can be measured by considering basin entropy and also considering stability in terms of cycle diversity. Thus, through evaluating these properties of different structural motifs, Ahnert and Fink (2016) argue that evaluating these properties could provide a shortcut for interpreting system-level dynamics.

Huang et al. (2022) validates some of the work of Ahnert and Fink (2016) and brings it further by providing a quantitative scoring method for defining networks. This work further defines network functionality as a balance of multiplicity and flexibility. Huang et al. (2022)

defines multiplicity as diverse behaviors for differentiation and explains that multiplicity is the GRN's ability to generate rich dynamical behaviors. Its flexibility is a measure of its responsiveness to the environment which is related to its ability to be controlled by extrinsic cell signaling or environmental factors (Huang et al., 2022). Huang et al. (2022) identifies motifs with sparse connectivity and self-activation to be among the network motifs with both high multiplicity and high flexibility. Additionally, motifs with monodirectional regulation and without autoregulation tend to have high flexibility and not high multiplicity. To better understand how this might impact our work, it is also often difficult to have multiplicity in small GRNs because when a GRN is too small it is not complex enough to generate desired functionality and thus our small networks might have limited functionality. Thus, we can look to see which of our networks would be more likely to have high flexibility. While none of the networks we studied have strictly monodirectional regulation, the two networks shown in Figures 7a and 7b do not have autoregulation. We can infer that the performance of network 7 (Figure 7b) is the best over the two other networks because it is more flexible. Comparing the network originally ranked first to the network originally ranked seventh we see that the network originally ranked seventh is more monodirectional than the network originally ranked first. Thus, it would be interesting to continue this work and rank the networks based on flexibility to see if that correlates to their accuracy. Additionally, we could use these properties to validate our original networks and see how our analysis aligns with the multiplicity and flexibility of DB1, DB2, and DB5 as a future pathway (Huang et al., 2022).

While the current work is comprehensive in that results from 147 model runs were compiled and analyzed, there are still some limitations in the study. For the sensitivity analysis of the literature-derived GRNs, it would be helpful to investigate the robustness of the parameter

estimation for the other networks, DB3, DB4, DB6, and DB7 to gain additional insights as to which genes and network structures contributed to robustness versus instability in estimated parameter values. For the 3-gene, 4-edge networks, it would be helpful to be able to definitively rank the network structures by modeling all different permutations of each of the original 21 networks. Even for the three networks for which the permutation study was performed, we only looked in detail at the results when all three parameters were estimated and when only w was estimated. It is also necessary to continue the sensitivity analysis by determining how the network performs when noise is systematically added back into the simulated gene expression data. At this time, we do not know if our current conclusions will continue to be valid when noise is added to the data. Additionally, it would be interesting to trim down one of the literature-derived networks we have been working with to see if the motifs present in that network perform in a similar way when experimental expression data is used compared to the motifs we have studied with simulated data. Lastly, because of the importance of production rates to the performance of the model throughout all of our experiments, it would be helpful to measure production rates for each gene in yeast under cold shock conditions so that experimentally derived values can be used as input to the model.

In conclusion, through first exploring the differences in parameter estimation in literature-derived yeast GRNs when expression data with missing datapoints and when these datapoints were replaced with the average value for that gene, strain, and timepoint and no missing data were input to the model, we identified the need to supply experimentally derived production rates to improve parameter stability. Despite changing the alpha value, discrepancies between the model outputs persisted which led us to begin a sensitivity analysis where we

explored all possible 3-gene, 4 edge networks. Through this work we concluded that both network weights and network structure impact weight estimation accuracy.

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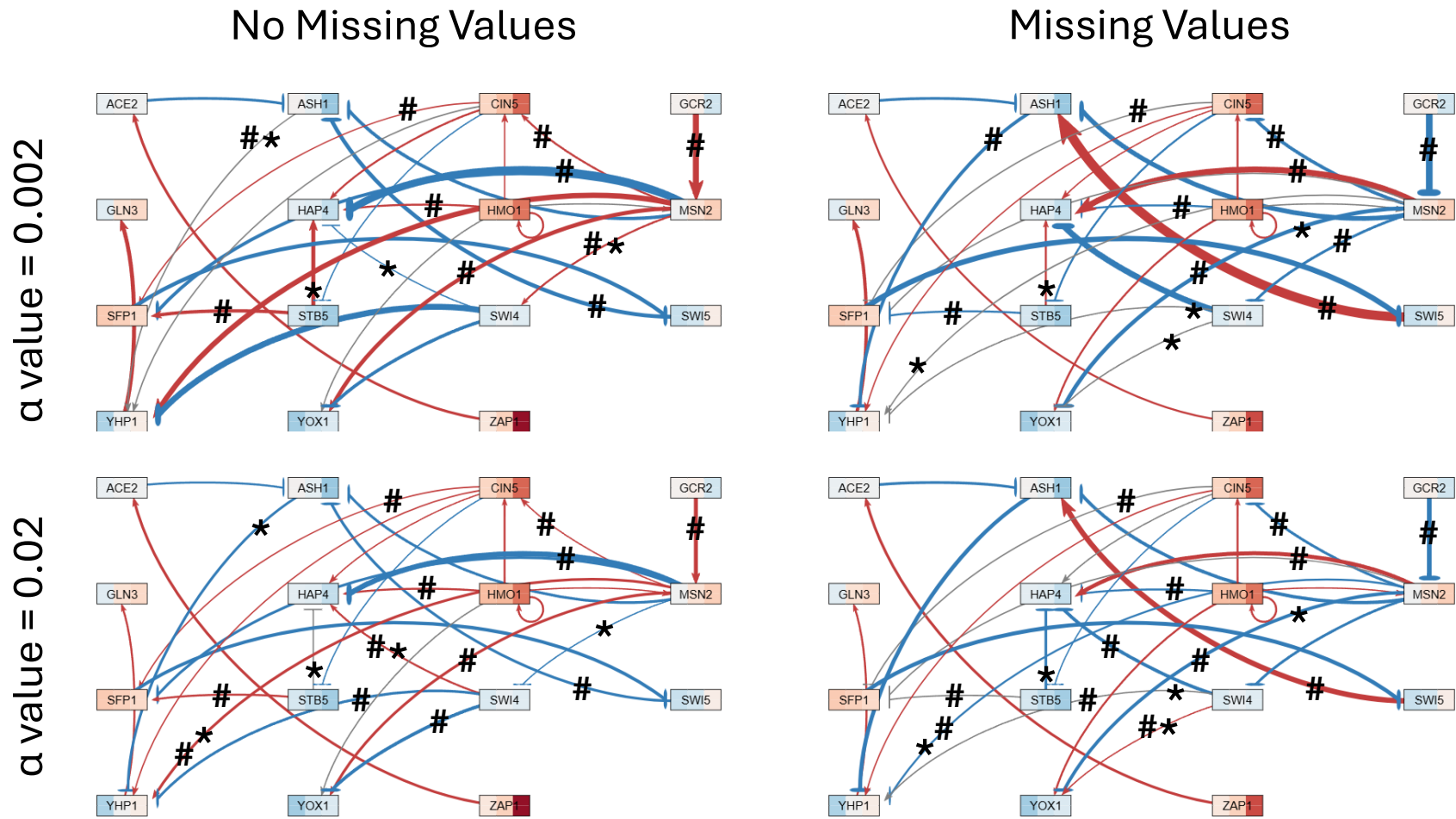


Figure 1. Weighted GRN of DB5 as visualized with GRNsight. Model results from workbooks where missing expression data values were replaced with the average value for that gene and timepoint (“no missing values”, left) compared to results from workbooks with missing expression values (“missing values”, right) at two different alpha values, 0.002 (top) and 0.02 (bottom).. Node coloring depicts experimentally measured wild type log₂ expression at 15, 30, and 60 minutes of cold shock with the log fold change max value set to 3 for color intensity where red is an increase in expression and blue is a decrease in expression. Blue, blunt arrows indicate repression and red, pointed arrows indicate activation with thicker arrows representing a larger magnitude weight. The edge weight normalization factor was set to 5, and edges with weights within 5% of zero are shown in grey. Asterisks (*) indicate weight values which changed sign between the different alpha value outputs and (#) indicate weight values that changed sign between missing and no missing values.

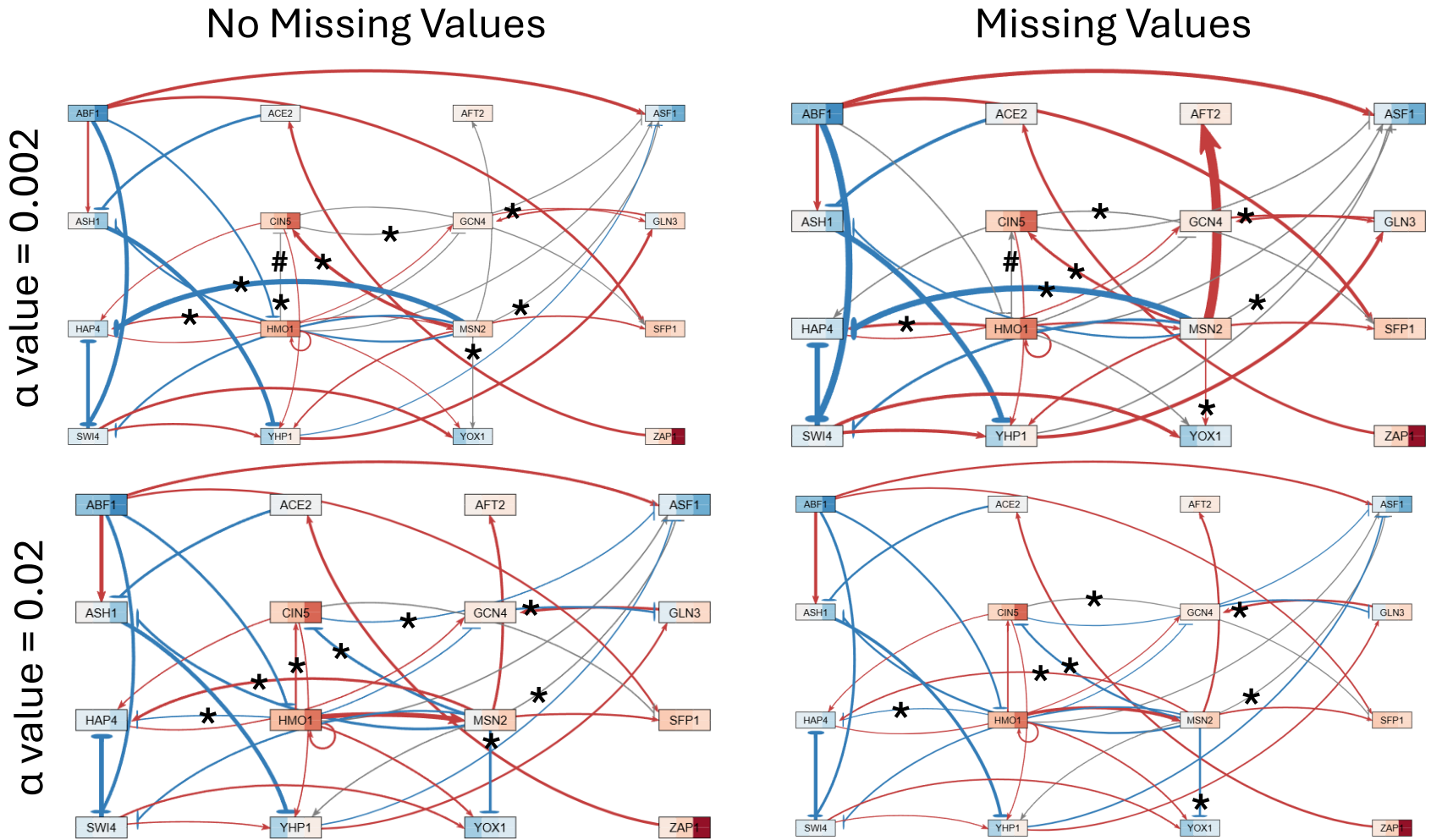


Figure 2. Weighted GRN of DB1 as visualized with GRNsight. Model results from workbooks where missing expression data values were replaced with the average value for that gene and timepoint (“no missing values”, left) compared to results from workbooks with missing expression values (“missing values”, right) at two different alpha values, 0.002 (top) and 0.02 (bottom).. Node coloring depicts experimentally measured wild type $\log_2$ expression at 15, 30, and 60 minutes of cold shock with the log fold change max value set to 3 for color intensity where red is an increase in expression and blue is a decrease in expression. Blue, blunt arrows indicate repression and red, pointed arrows indicate activation with thicker arrows representing a larger magnitude weight. The edge weight normalization factor was set to 5, and edges with weights within 5% of zero are shown in grey. Asterisks (*) indicate weight values which changed sign between the different alpha value outputs and (#) indicate weight values that changed sign between missing and no missing values.

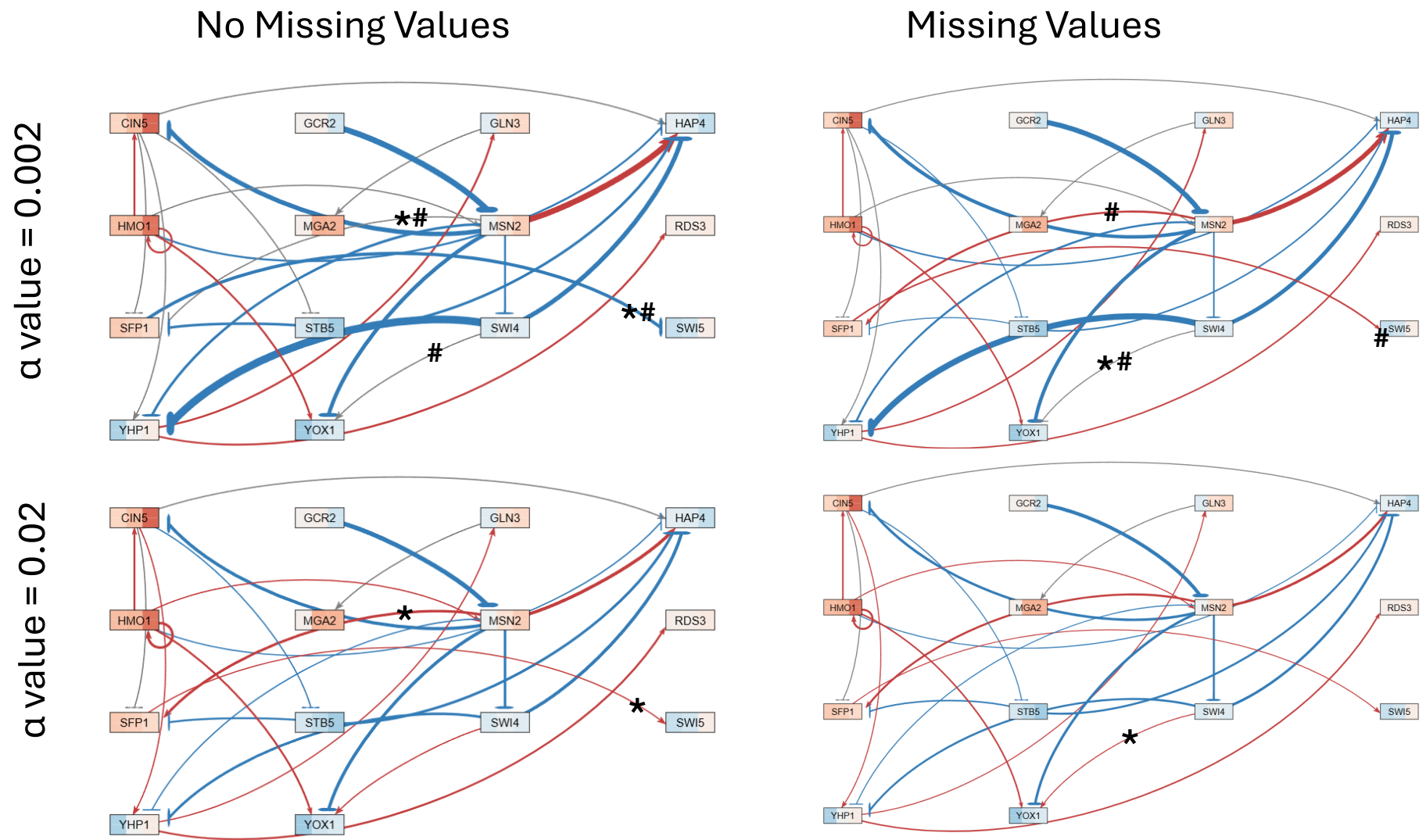


Figure 3. Weighted GRN of DB2 as visualized with GRNsight. Model results from workbooks where missing expression data values were replaced with the average value for that gene and timepoint (“no missing values”, left) compared to results from workbooks with missing expression values (“missing values”, right) at two different alpha values, 0.002 (top) and 0.02 (bottom).. Node coloring depicts experimentally measured wild type log₂ expression at 15, 30, and 60 minutes of cold shock with the log fold change max value set to 3 for color intensity where red is an increase in expression and blue is a decrease in expression. Blue, blunt arrows indicate repression and red, pointed arrows indicate activation with thicker arrows representing a larger magnitude weight. The edge weight normalization factor was set to 5, and edges with weights within 5% of zero are shown in grey. Asterisks (*) indicate weight values which changed sign between the different alpha value outputs and (#) indicate weight values that changed sign between missing and no missing values.

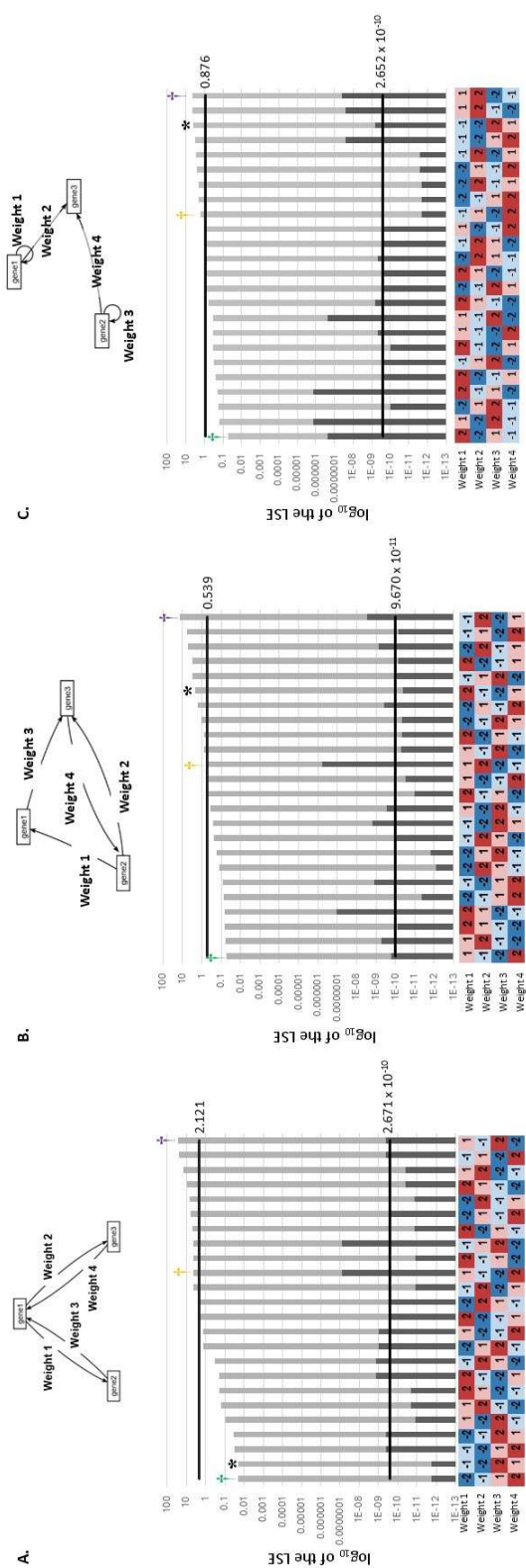


Figure 4. Log₁₀ weight LSEs for the networks ranked 1st (A), 7th (B), and 19th (C) when weight parameter were permuted. The dark bars represent when only w was estimated and the lighter grey bars represent when P , b , and w were all estimated, sorted by the lighter bars. Below each bar is a scheme showing input weights for each network. The values of each weight are shown: -2 (dark blue), -1 (light blue), 1 (pink), and 2 (red). *indicates the network configuration for the models shown in Figure 4 and Table 13. The †corresponds to Fig. 8. Each horizontal line represents the median LSE of each condition.

Appendix B

The raw data and the summary files used to create all of the figures and data are available on the Dahlquist Lab GitHub repository. The data is organized by figure/table, and each folder contains a ‘readme.md’ file which explains the contents of the folder if more explanation is needed. The thesis folder is available [here](#), and the following links are organized by figure in the order they appear in the text.

Figure 1: <https://github.com/kdahlquist/DahlquistLab/tree/master/data/Spring2026/Thesis/Figure%201>
Table 1: <https://github.com/kdahlquist/DahlquistLab/tree/master/data/Spring2026/Thesis/Table%201>
Table 2: <https://github.com/kdahlquist/DahlquistLab/tree/master/data/Spring2026/Thesis/Table%202-4>
Table 3: <https://github.com/kdahlquist/DahlquistLab/tree/master/data/Spring2026/Thesis/Table%202-4>
Table 4: <https://github.com/kdahlquist/DahlquistLab/tree/master/data/Spring2026/Thesis/Table%202-4>
Figure 2: <https://github.com/kdahlquist/DahlquistLab/tree/master/data/Spring2026/Thesis/Figure%202>
Table 5: <https://github.com/kdahlquist/DahlquistLab/tree/master/data/Spring2026/Thesis/Table%205-7>
Table 6: <https://github.com/kdahlquist/DahlquistLab/tree/master/data/Spring2026/Thesis/Table%205-7>
Table 7: <https://github.com/kdahlquist/DahlquistLab/tree/master/data/Spring2026/Thesis/Table%205-7>
Table 8: <https://github.com/kdahlquist/DahlquistLab/tree/master/data/Spring2026/Thesis/Table%208-10>
Table 9: <https://github.com/kdahlquist/DahlquistLab/tree/master/data/Spring2026/Thesis/Table%208-10>
Table 10: <https://github.com/kdahlquist/DahlquistLab/tree/master/data/Spring2026/Thesis/Table%208-10>
Table 11: <https://github.com/kdahlquist/DahlquistLab/tree/master/data/Spring2026/Thesis/Table%2011>
Table 12: <https://github.com/kdahlquist/DahlquistLab/tree/master/data/Spring2026/Thesis/Table%2012>
Figure 3: <https://github.com/kdahlquist/DahlquistLab/tree/master/data/Spring2026/Thesis/Figure%203>
Figure 4: <https://github.com/kdahlquist/DahlquistLab/tree/master/data/Spring2026/Thesis/Figure%204>
Table 13: <https://github.com/kdahlquist/DahlquistLab/tree/master/data/Spring2026/Thesis/Table%2013>
Figure 5: <https://github.com/kdahlquist/DahlquistLab/tree/master/data/Spring2026/Thesis/Figure%205>
Figure 6: <https://github.com/kdahlquist/DahlquistLab/tree/master/data/Spring2026/Thesis/Figure%206>
Figure 7: <https://github.com/kdahlquist/DahlquistLab/tree/master/data/Spring2026/Thesis/Figure%207>
Figure 8: <https://github.com/kdahlquist/DahlquistLab/tree/master/data/Spring2026/Thesis/Figure%208>
Appendix A Figure 1: <https://github.com/kdahlquist/DahlquistLab/tree/master/data/Spring2026/Thesis/Appendix/Fig%201>
Appendix A Figure 2: <https://github.com/kdahlquist/DahlquistLab/tree/master/data/Spring2026/Thesis/Appendix/Fig%202>
Appendix A Figure 3: <https://github.com/kdahlquist/DahlquistLab/tree/master/data/Spring2026/Thesis/Appendix/Fig%203>
Appendix A Figure 4: <https://github.com/kdahlquist/DahlquistLab/tree/master/data/Spring2026/Thesis/Appendix/Fig%204b>